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Ethanol liquid film-assisted nanosecond laser transmission welding for stainless steel and glass

Runhan Zhao ^a, Yide Li ^b, Jingyi Luo ^a, Zhijiao Liu ^a, Jiajun Fu ^a, Chao Liu ^a, Huixin Wang ^{c, d}, Qinghua Wang ^a , Tairui Zhang ^a [Show more](#) ▼

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Highlights

- A liquid film enables fixture-free nanosecond laser welding of glass and metal.
- Straight-line welding with frame and interior filling improves joint performance.
- The shear strength of 19.73MPa is achieved using an optimized scanning strategy.
- Joint evolution and fracture models are analyzed for welding mechanism insights.

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Keywords

Nanosecond laser; Laser welding; Dissimilar materials; Mechanical strength

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1. Introduction

The stable joining of dissimilar materials such as glass and metal plays a pivotal role in various fields, including electronics, optics, and medical devices [[1], [2], [3]]. However, the substantial differences in their physical and chemical properties, particularly the mismatch in their coefficients of thermal expansion [4], pose considerable difficulties in establishing reliable and efficient connections. Traditional methods for joining glass and metals, such as adhesive bonding [5], brazing [6], electron beam welding [7], and arc welding [8], often suffer from limitations like low precision, inadequate joint strength, and the tendency to induce substantial welding stresses. These conventional methods do not align with the evolving needs of modern manufacturing, which increasingly requires high-precision and robust connections for complex structures.

Laser welding, based on laser-material interactions, has emerged as a promising method for joining dissimilar materials due to its high precision, non-contact processing, minimal heat-affected zone (HAZ), and adaptability to a wide range of materials [9]. This technique offers a potential solution to the long-standing issues in joining glass and metals. Laser welding typically employs a continuous or pulsed laser as the energy source, focusing the laser beam onto the sample through a lens. The laser energy induces both linear and nonlinear absorption, facilitating materials fusion. Laser transmission welding (LTW) has been widely adopted for joining transparent materials, such as glass

and plastic [10]. Compared to traditional methods, laser welding enables highly precise operations at the micron and nanometer scales, offering advantages in welding efficiency, cost-effectiveness, and reliability. Nevertheless, thermal stress issues persist, influencing the overall quality of the joint.

To precisely control the thermal effects during welding of dissimilar materials with substantial differences in their coefficients of thermal expansion, ultrafast lasers (including picosecond and femtosecond lasers) have become increasingly popular for high-precision welding of brittle and transparent materials [11]. These lasers produce extremely short pulse durations and very small HAZs, which localize energy through transient optical nonlinear effects. This significantly reduces the risk of defects caused by thermal stress concentration and enables high-precision welding. In 2005, Tamaki et al. [12] successfully welded glass to monocrystalline silicon with a weld strength of 3.74MPa, using a high repetition rate femtosecond fiber laser that relied on a thermal cumulative effect. That same year, Watanabe et al. [13] used a femtosecond laser with a pulse repetition frequency of 1kHz to weld dissimilar materials through laser surface scanning, and a weld strength of 15.3MPa was attained. In 2008, Alexander et al. [14] successfully welded glass and semiconductor materials with a 700kHz femtosecond laser and observed that material fusion occurred in the weld zone, where silicon was mixed into the glass. Huan et al. [15] discovered that femtosecond lasers with high repetition rates and low pulse energy effectively inhibited the formation of brittle intermetallic phases and reduced welding defects. Pan et al. [16] reported a weld strength of 108.35MPa when joining sapphire and Fe—36Ni alloy directly through femtosecond laser micro-welding, with the materials chemically bonded and mechanically interlocked. Similarly, picosecond lasers have also been extensively studied for welding dissimilar materials. Compared to femtosecond lasers, they offer higher efficiency for welding transparent materials. Carter et al. [17] employed a picosecond laser to weld glass with stainless steel through helical welding, while microcracks (100–200μm) were observed in the seam. Jia et al. [18] proposed a method for reliably bonding copper-plated glass to metal through continuous-wave picosecond laser welding. Their results indicated that the increased

laser power and irradiation time enhanced weld strength, with a maximum strength of 39.17MPa. Ciuca et al. [19] effectively welded fused silica glass to aluminum using picosecond pulsed lasers, attaining a weld strength of 26MPa. Despite some advantages, the high cost, complexity of equipment, and stringent environmental and operational requirements of ultrafast lasers limit their large-scale industrial applications.

In contrast, nanosecond laser exhibits some distinct advantages, including compactness, cost-effectiveness, ease of maintenance, stability, and acceptable thermal impact [20]. Nanosecond laser welding has shown significant potential in joining dissimilar materials, particularly highly absorbing substrates such as metals and opaque ceramics. In 2007, Utsumi et al. [21] used a nanosecond pulsed laser to join borosilicate glass to a copper sphere directly, while the resulting strength was only 0.87MPa. Zhang et al. [22] applied a nanosecond pulsed laser to join YSZ to sapphire under optical contact conditions with a weld strength of 32MPa and systematically investigated the organization, mechanical properties, and forming process of the YSZ/Sapphire joint. Ji et al. [23] fabricated pit microstructures with a spacing of 100 μ m on the bonding surface of a Ti₆Al₄V alloy and ZrO₂ ceramic under optical contact conditions, resulting in a maximum strength of 46MPa. Huo et al. [24] proposed a nanosecond laser welding method for glass and stainless steel by utilizing an intermediate layer of chromium ion solution. Elemental diffusion and interfacial reactions were observed with a maximum strength of 41.4MPa. Wang et al. [25] introduced a 30nm titanium coating as an absorbing layer between two pieces of glass and welded them using a nanosecond laser. They systematically investigated the effects of processing parameters, including laser power, scanning speed, and repetition frequency, on the transmittance and tensile strength of the joint, and a maximum weld strength of 38.07MPa was obtained. Tian et al. [26] combined nanosecond laser welding with picosecond ultrafast laser welding, capable of connecting glass and metal with a gap of a few hundred micrometers. However, the device design is complex and requires the coordinated operation of two lasers.

These studies illustrate the feasibility of using nanosecond lasers for dissimilar material welding. Nanosecond lasers induce a plasma plume at the weld interface with a longer

duration of action, utilizing heat buildup to mix the molten materials and achieving a stable joint [22]. However, these welding methods typically require additional ionic solutions or metallic interlayers to ensure a high-strength and stable connection. Without these materials, external pressure is generally required to maintain optical contact between the workpieces (with a sample gap of less than $\lambda/4$) in order to achieve stable weld strength [27]. Optical contact requires precise alignment and stable optical transmission to ensure that the plasma generated by the laser remains contained, which allows for the welding of dissimilar materials. If the gap between the dissimilar materials is too large, the melted material will fail to fill the gap, resulting in a low-quality joint [[28], [29], [30]]. This reliance on optical contact complicates the welding process, limits the welding range, and makes precise pressure quantification difficult, which will reduce welding quality and productivity. Thus, the development of a cost-effective, simple, and efficient nanosecond laser welding process is of great scientific and practical importance for advancing dissimilar material welding technology.

In summary, to address the obstacles of the narrow processing parameter range of nanosecond laser welding and the difficulty in the realization of optical contact, this study introduces a liquid film between silica glass and stainless steel for the laser welding process. The surface tension of the liquid film replaces the need for conventional external pressure, ensuring optical contact between the workpieces and contributing to the formation of a reliable solder joint. Despite extensive studies on welding path design and interface optimization in ultrafast and nanosecond laser welding, few have explored the use of liquid interlayers for pressure-free welding of glass-to-metal systems. Additionally, there is limited investigation into how different scanning strategies affect the interface bonding quality, especially under fixture-free and intermediate-layer-free conditions. In this work, we systematically examine two representative laser welding patterns [15,31], comparing concentric circles and straight lines as the fundamental welding elements. In the meantime, three welding strategies were explored in detail to assess their influence on bonding performance and weld morphology: welding the outer frame only (SL-strategy 1), filling the interior only (SL-strategy 2), and combining the outer frame with

interior filling (SL-strategy 3). Additionally, the differences between liquid film-assisted transmission welding (LFTW) and external pressurized welding are systematically evaluated from the three-dimensional (3D) morphology, and microstructure. The role of surface tension in the welding process, the advantages and disadvantages of the two methods, and the different fracture modes of the joint are analyzed in depth. This study aims to establish a simplified, low-cost, and pressure-free laser welding strategy suitable for industrial implementation in the joining of dissimilar materials.

2. Experiment details

2.1. Materials

In this study, silica glass with a purity of 99.9% ($\phi 20\text{ mm} \times 2\text{ mm}$) and 304 mirror-finish stainless steel ($35\text{ mm} \times 35\text{ mm} \times 0.7\text{ mm}$) were selected as the experimental materials. Both the stainless steel and glass surfaces were polished, with the surface roughness (R_a) of the stainless steel approximately $25\text{ }\mu\text{m}$ and that of the glass approximately 70 nm . The thermophysical properties of the glass and stainless steel, as well as the components of the 304 stainless steel, are provided in [Table 1](#) and [Table 2](#), respectively. Notably, the silica glass exhibits a transmittance of over 99% for red light.

Table 1. The thermophysical properties of stainless steel and glass.

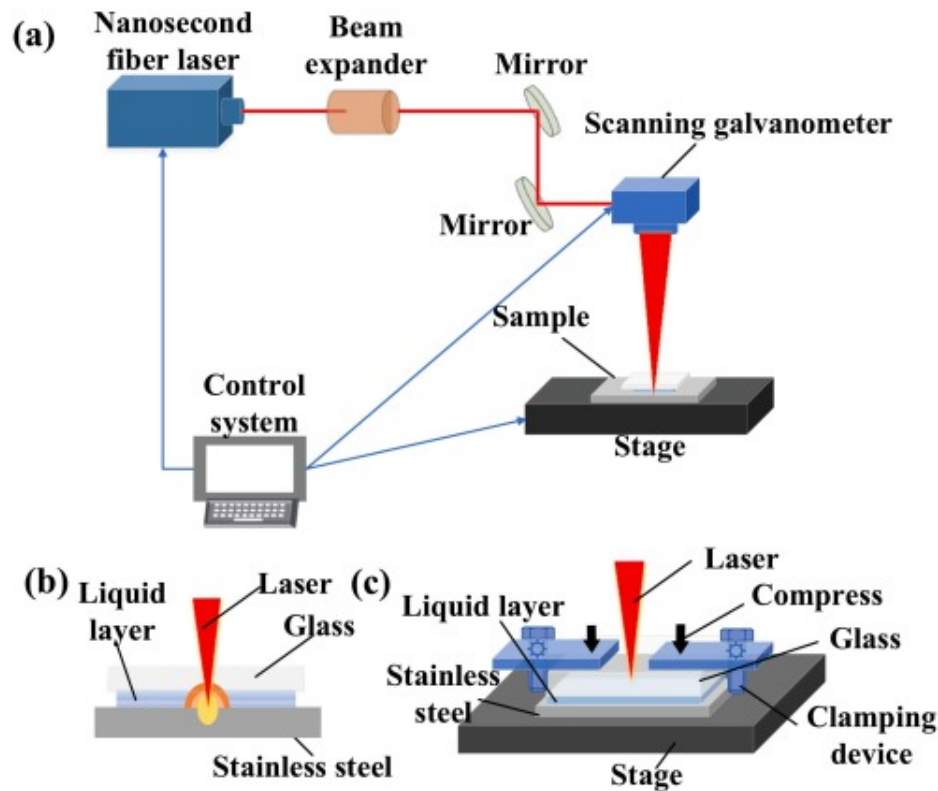
Material	T_{Melt} ($^{\circ}\text{C}$)	σ_{Tensile} (MPa)	Coefficient of linear expansion	Thermal diffusivity (mm^2/s)	Thermal conductivity ($\text{W}/(\text{m}\cdot\text{K})$)
Stainless steel	1440	520	17.2×10^{-6}	3.835	14.986
SiO_2 glass	1723	40–70	5.5×10^{-7}	1.3	0.95

Table 2. The weight percentage (wt%) of various components in stainless steel.

Component	C	Mn	Si	P	S	Ni	Cr	N	Fe
wt (%)	0.049	0.93	0.44	0.028	0.004	8.00	18.03	0.062	72.457

2.2. Nanosecond laser equipment and welding process

The laser welding system is depicted in Fig. 1(a). This system comprises a nanosecond fiber laser, a beam expander, mirrors, and a scanning galvanometer. The nanosecond fiber laser (YDFLP-E-60-M7-M-R, JPT, Shenzhen) operates at a wavelength of 1064nm, with a fixed pulse duration of 200ns. The spot diameter at the focal point is approximately 60μm. Fig. 1(b) presents a schematic of the LFTW process, wherein the sample consists of glass, an intermediate liquid film, and a stainless steel substrate. The laser beam is focused on the contact interface between the glass and the stainless steel.

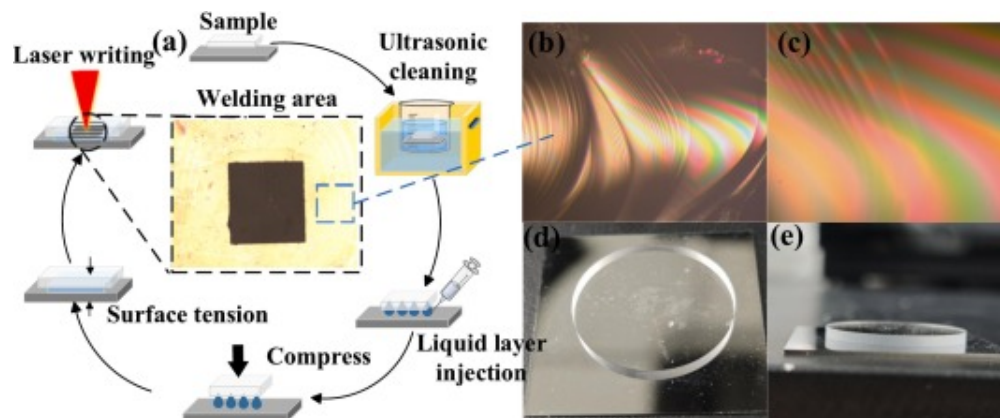


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Fig. 1. (a) Is a schematic diagram of the laser welding system, (b) depicts the LFTW process, and (c) illustrates the external pressurization-assisted treatment method.

Pretreatment is required to ensure optimal optical contact between the glass and stainless steel, as detailed in Fig. 2. Additionally, Fig. 1(c) illustrates the external pressurization-assisted method. This method involves fixing the sample (prepared as shown in Fig. 1(b)) under a spacer secured with bolts and nuts and then applying external pressure to enhance the welding effect.



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Fig. 2. (a) Is a flowchart of the pretreatment process of the LFTW method; (b) and (c) are the microscope images of the Newton rings observed after welding, and (d) and (e) are images of the pretreated samples.

Several types of injected liquids were tested in preliminary experiments, including distilled water, metal ion solutions, and anhydrous ethanol. Among these, ethanol demonstrated the best performance in maintaining optical contact and minimizing interference with subsequent laser transmission. Its fast volatility ensures that no residue remains after initial irradiation. In contrast, distilled water evaporates slowly and tends to accumulate at the weld interface, resulting in lower shear strength. Although metal ion solutions could offer enhanced absorption, limited concentration and slow evaporation rendered them less effective in our configuration. Therefore, ethanol was selected for its superior welding compatibility.

Fig. 2(a) illustrates the detailed pretreatment process. Initially, a polished mirror-finish stainless steel piece and a silica glass sample were selected, respectively. The glass and 304 stainless steel samples were then ultrasonically cleaned in anhydrous ethanol for 5 min to remove surface contaminants, followed by drying with absorbent paper. A

pipette was used to inject approximately 0.5 μL of anhydrous ethanol to form a liquid film between the glass and stainless steel. Pressure was subsequently applied to compress the gap and expel any excess liquid around the glass. The specific value of the applied pressure was quantitatively measured using the flexible thin-film pressure sensor, as shown in Fig. 4(c). The measurements revealed that the external pressure during the ethanol film formation process ranged from approximately 3 N to 5 N, achieved by manually pressing the glass for a duration of about 10 s. After the external pressure was released, the samples were maintained in optical contact (gap <200 nm) through surface tension and Vander Waals force [32]. At this point, the laser was focused on the surface of the stainless steel substrate to initiate the welding. The side views of the pretreated sample are shown in Fig. 2(d) and Fig. 2(e). After welding, the samples were examined under an optical microscope, as depicted in the middle of Fig. 2(a) and in Fig. 2(b) and Fig. 2(c), which illustrates the morphology of the welded area. In Fig. 2(a), Newton rings, formed due to the optical contact, are visible around the welded area, with the magnified views in Fig. 2(b) and Fig. 2(c) clearly showing these rings.

2.3. Experiment design

2.3.1. Comparative exploration of welding element and scanning path strategy

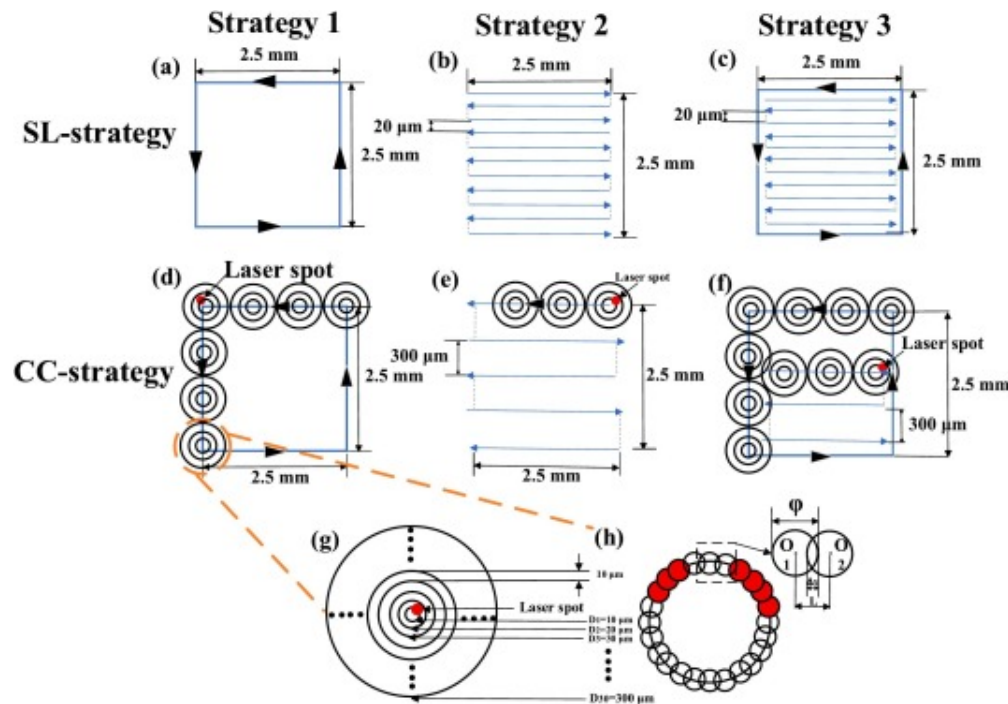
In this study, comparative experiments were conducted using six groups of parameters, as listed in Table 3. The parameters were varied over a wide range by adjusting overlap and power, with the goal of identifying the optimal scanning path strategy and corresponding parameters for achieving a stable weld pattern. The selected range of process parameters (laser power, scanning speed, repetition frequency) was derived from a series of pre-experiments. The parameter window was designed to cover a broad and ascending heat input spectrum, enabling analysis of both under-exposure and over-exposure conditions. Higher parameter values result in increased thermal accumulation and glass surface ablation [33] while lower values result in poor bonding or no joint formation due to

insufficient melting at the interface. Therefore, the final values listed in Table 3 represent the effective operating window for successful ethanol-assisted transmission welding of dissimilar materials. The formula for the calculation of overlap is provided in Eq. (1). Since welding scanning patterns typically utilize concentric circles and straight lines [31] as fundamental elements, this study compares these two welding elements using three distinct scanning path strategies, as illustrated in Fig. 3. Additionally, anhydrous ethanol was employed as an intermediate liquid layer to facilitate bonding between the glass and stainless steel substrate through the LFTW method.

$$S_{overlap} = \varphi = \frac{\Delta x}{\Delta d} = 1 - \frac{v_{scan}}{f_{rep} \times d_f} \quad (1)$$

Table 3. Parameters utilized in nanosecond laser welding.

Parameters	Group 1	Group 2	Group 3	Group 4	Group 5	Group 6
Pulse energy (W)	1.8	2.4	2.4	3	6	18
Repetition frequency (kHz)	20	30	35	40	60	80
Scanning speed (mm/s)	80	70	65	60	50	40
Scanning times (times)	50	60	65	70	80	90
S _{overlap}	86.6%	92.2%	93.8%	95%	97.2%	98.3%



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Fig. 3. A schematic comparing two types of welding elements, concentric circles and straight lines, employing three different scanning path strategies.

In Eq. (1), $S_{overlap}$ represents the spot overlap, v_{scan} denotes the scanning speed (mm/s), d_f is the spot diameter after focusing (μm), and f_{rep} is the repetition frequency (kHz).

The scanning area of the weld region, as shown in Fig. 3, is fixed at $2.5\text{ mm} \times 2.5\text{ mm}$. To systematically evaluate the effect of path design on weld formation, three scanning strategies were compared: scanning only the outer frame (Strategy 1), filling the interior only (Strategy 2), and a combined two-step strategy of first scanning the outer frame and then filling the interior (Strategy 3). Strategy 1 was designed to explore the effect of boundary stabilization alone, while Strategy 2 focused on direct interior bonding without edge constraint. Strategy 3 aimed to combine the advantages of both to achieve

maximum bonding strength and morphological stability. The purpose of this comparison was to validate the effectiveness of Strategy 3 in stabilizing the weld boundary and maintaining optical contact during the entire welding process. Fig. 3(a)-(c) illustrate the schematic diagrams for these three strategies using straight lines as the fundamental welding element, abbreviated as SL-strategy 1–3. Similarly, Fig. 3(d)-(f) present the schematic diagrams for the strategies using concentric circles as the fundamental welding element, abbreviated as CC-strategy 1–3. Fig. 3(g) details the specific concentric circle scanning paths, where multiple concentric circle trajectories are employed, progressing from the inside to the outside. Given that the spot size at the laser focus is approximately $60\mu\text{m}$, the scanning path begins with the inner concentric circle (diameter of $10\mu\text{m}$, scanning spacing of $5\mu\text{m}$) and extends outward to the outer concentric circle (diameter of $300\mu\text{m}$). The beam scans along this planned path, removing material layer by layer [34]. Notably, the scanning spacing for SL-strategy 2 and SL-strategy 3 is $20\mu\text{m}$. In contrast, for CC-strategy 2 and CC-strategy 3, the scanning spacing corresponds to the diameter of the outermost circle in the concentric paths ($300\mu\text{m}$), effectively preventing overlap between concentric circles.

2.3.2. Comparative experiments under different operating conditions

Following the comparative study of welding patterns and scanning strategies, the optimal welding element and scanning path strategy were selected for subsequent experiments. During welding, the thermal expansion of materials due to laser heating can increase the gap between the glass and stainless steel substrate, potentially causing welding failure. Therefore, it is essential to ensure optical contact before welding and maintain a minimal gap throughout the process. To evaluate the effectiveness of the LFTW method, external pressurization was introduced as a comparison in this section. External pressurization was applied to further minimize the gap, maintain consistent optical contact, and evaluate changes in weld strength. Comparative experiments were conducted under four distinct operating conditions, using the optimal parameters identified in Section 2.3.1. The detailed operating conditions and parameters are presented in Table 4. The four operating conditions tested in this section are as follows: No intermediate layer, no external

pressure (NIL-NEP); Alcohol liquid as the intermediate layer, no external pressure (AL-NEP); No intermediate layer, with external pressure (NIL-EP); Alcohol liquid as the intermediate layer, with external pressure (AL-EP). To ensure the stability of the welding, each experiment was conducted three times under the same operating condition, and the average weld strength was calculated. Furthermore, the welded area was maintained at 2.5 mm×2.5 mm, consistent with [Section 2.3.1](#).

Table 4. The parameters and operating conditions utilized in [Section 2.3.2](#).

Laser parameters	Operating conditions
Pulse energy: 3W	No intermediate layer, no external pressure (NIL-NEP)
Repetition frequency: 40kHz	Alcohol liquid as the intermediate layer, no external pressure (AL-NEP)
Scanning speed: 60mm/s	No intermediate layer, with external pressure (NIL-EP)
Scanning times: 70 times	Alcohol liquid as the intermediate layer, with external pressure (AL-EP)

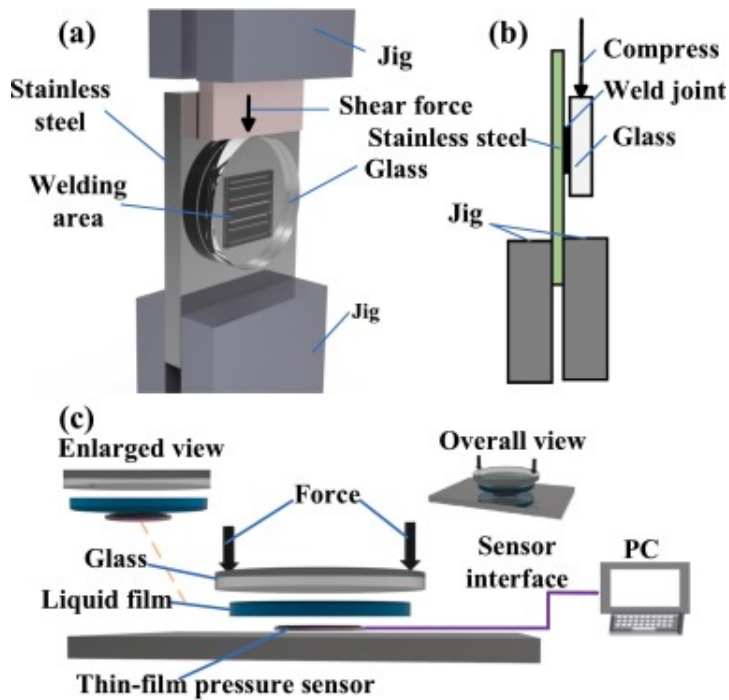
2.4. Characterization and measurement methods

Shear strength serves as an indicator of weld strength. The shear strength of the welded samples was measured to evaluate the weld quality using a digital tensile tester (AILIGU, Shenzhen). [Fig. 4\(a\)](#) and [Fig. 4\(b\)](#) illustrate the schematic representation of the testing procedure. During the test, the welded workpiece was secured on the lower fixture, and an indenter applied downward pressure to the glass parallel to its edge. The critical thrust force at which the stainless steel was separated from the glass was recorded. It is essential to ensure that the contact point of the applied pressure aligns vertically with the center of the welded area to eliminate any bending moment interference on the test results. The

shear strength (σ) was calculated using Eq. (2).

$$\sigma = \frac{F}{S} \quad (2)$$

where F denotes the applied shear force and S represents the weld area. To minimize measurement errors, each measurement was repeated three times, and the average value was recorded.



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Fig. 4. (a) 3D schematic of shear strength measurement; (b) 2D schematic diagram of shear strength measurement; (c) Schematic of flexible thin-film sensor, placed between glass and stainless steel plate.

An optical microscope (UMM-IF-5) was employed to examine the appearance of the welded area, providing a preliminary assessment of the weld morphology and the HAZ

distribution. Following shear strength testing of the welded samples, the two-dimensional (2D) and 3D morphologies of the joint interface between the fractured glass and stainless steel were meticulously characterized using a digital stereomicroscope (VHX-6000, Keyence). The microstructure of the fractured surfaces was analyzed using field emission scanning electron microscopy (SEM, Sirion, FEI), and the elemental composition of the microstructure was examined using an energy-dispersive spectroscopy (EDS) system (Sirion, FEI). For the characterization of the intact welded joint, the welded samples were embedded in an inlay solution and cut at the joint using a diamond wire-cutting machine (Metkon, Micracut 201). The intact welds were observed with a SEM (ZEISS, Sigma 360), and their elemental composition was analyzed through line and surface scanning using an EDS (OXFORD Xplore 30).

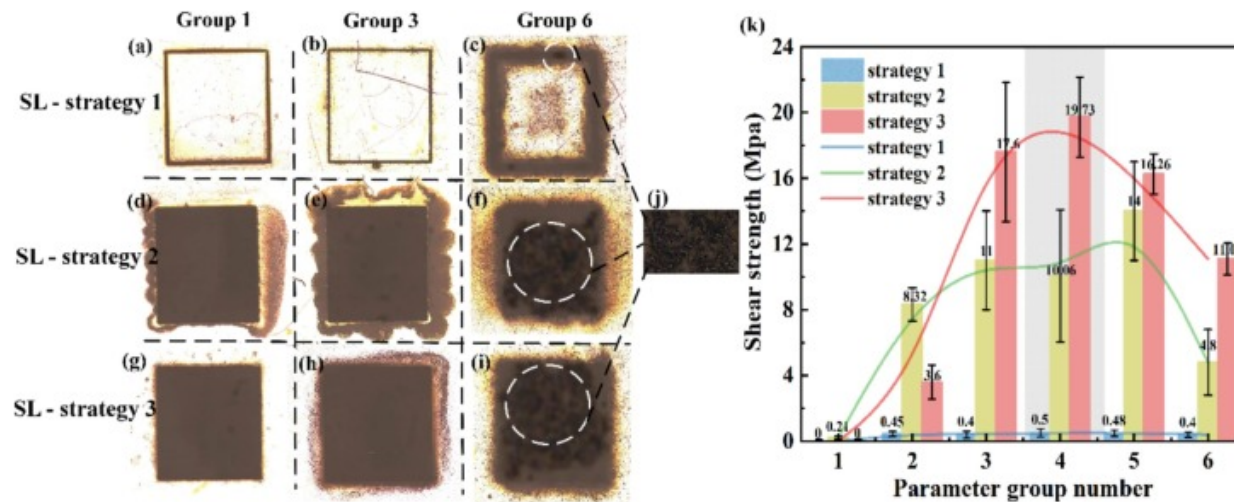
To accurately quantify both the pressure exerted during sample preparation and the actual pressure applied under externally loaded conditions, a flexible thin-film pressure sensor (RunesKee) was strategically integrated beneath the ethanol layer within the liquid-assisted laser welding configuration. The sensor was interfaced with a PC to enable real-time monitoring of applied force and time. For the externally-pressurized samples, the same type of sensor was placed between the glass and stainless steel substrates. This allowed accurate measurement of the pressure introduced by external mechanical loading.

The measurement results showed that the average external pressure applied using the clamping fixture during AL-EP and NIL-EP conditions was approximately 65N. This pressure was sufficient to ensure reliable optical contact between the dissimilar materials throughout the welding process, without causing visible damage to the glass surface. The pressure readings were stable and reproducible across multiple tests, demonstrating the consistency and effectiveness of the external mechanical loading strategy.

3. Results and discussion

3.1. Comparison between welding elements and scanning path strategies

Fig. 5 (a), (d), and (g) shows the results of welding joints following strategies 1 to 3 with Group 1 parameters, respectively. Similarly, Fig. 5(b), (e), and (h) correspond to Group 3 parameters, and Fig. 5(c), (f), and (i) illustrate the results using Group 6 parameters. As the parameters of the three groups vary, the overlap of the weld area increases, leading to higher laser energy concentration. Comparing Fig. 5(a)-(c), 5(d)-(f), and 5(g)-(i), it is evident that the HAZ expands with increased energy input under the same welding strategy. When Group 6 parameters are applied, micro-perforations are observed on the glass surface, as shown in Fig. 5(g). This phenomenon is attributed to the excessive energy input, which concentrates thermal stresses onto the glass surface, causing micro-perforations due to the repeated action of the plasma [35]. The black region at the edge of the welded area represents the HAZ, primarily caused by laser ablation of the stainless steel and alcohol carbonization [36]. Although SL-strategy 1 effectively minimizes the HAZ across all scanning path strategies, it generally yields lower shear strength, which is related to the formed weld, and the results of the shear strength are shown in Fig. 5(k). A comparison of Fig. 5(d)-(f) and Fig. 5(g)-(i) reveals that the HAZ in SL-strategy 2 is almost identical to that in SL-strategy 3, both of which are more extensive than that in SL-strategy 1.



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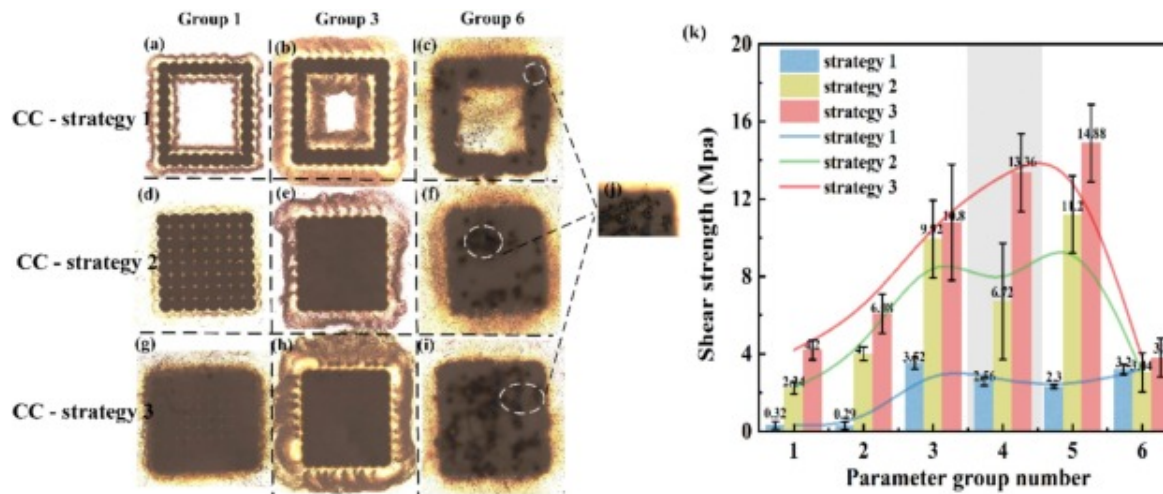
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Fig. 5. (a)–(c) Show the weld joints observed under an optical microscope using SL-strategy 1 with the parameters of Group 1, 3, and 6; (d)–(f) show the results using SL-strategy 2; (g)–(i) show the results using SL-strategy 3. (j) is an enlarged view of the surface micro-perforations on the glass corresponding to (c), (f), and (i). (k) illustrates the shear strength measurements for Groups 1–6 using SL-strategy 1–3. The curve represents the overall trend of welding strength. The gray background indicates that Group 4 parameters were selected as the baseline conditions for subsequent welding strategy comparison experiments.

The effects of the six groups of parameters on welding strength are shown in Fig. 5(k). As the $S_{overlap}$ and pulse energy increase, the heat accumulation effect becomes more pronounced, leading to an initial increase followed by a decrease in welding strength. The maximum shear strength of 19.73 MPa is attained with Group 4 parameters using SL-strategy 3. Excessive laser power leads to a decrease in shear strength due to the high thermal stress, causing cracks and defects in the glass. The experimental results indicate that none of the three scanning strategies yield practical welding results when Group 1

parameters are applied. With the parameters from Groups 2–6, a comparison of SL-strategy 1–3 reveals that SL-strategy 1 produces a weld with significantly lower strength. In contrast, SL-strategy 2 and SL-strategy 3 significantly enhance the welding strength, with SL-strategy 3 improving the weld performance by approximately 40% compared to SL-strategy 2. This highlights the substantial enhancement in weld quality with SL-strategy 3.

When concentric circles are used as the fundamental welding element, the results of surface morphology and welding strength are presented in Fig. 6. The results corresponding to Groups 1, 3, and 6, using CC-strategy 1–3, are shown in Fig. 6(a)(d)(g), Fig. 6(b)(e)(h), and Fig. 6(c)(f)(i), respectively. A comparison of the three groups of parameters reveals that as laser power and overlap increase, the HAZ enlarges significantly. Under Group 6 parameters, perforations on both the upper and lower surfaces of the glass are observed, as shown in Fig. 6(j), accompanied by a considerable increase microcracks. A comparison of CC-strategy 1–3 with SL-strategy 1–3 shows a markedly larger HAZ in the concentric circle strategies. This is because the diameter of the concentric circles is much larger than the line width of the straight lines, resulting in a more extensive HAZ.



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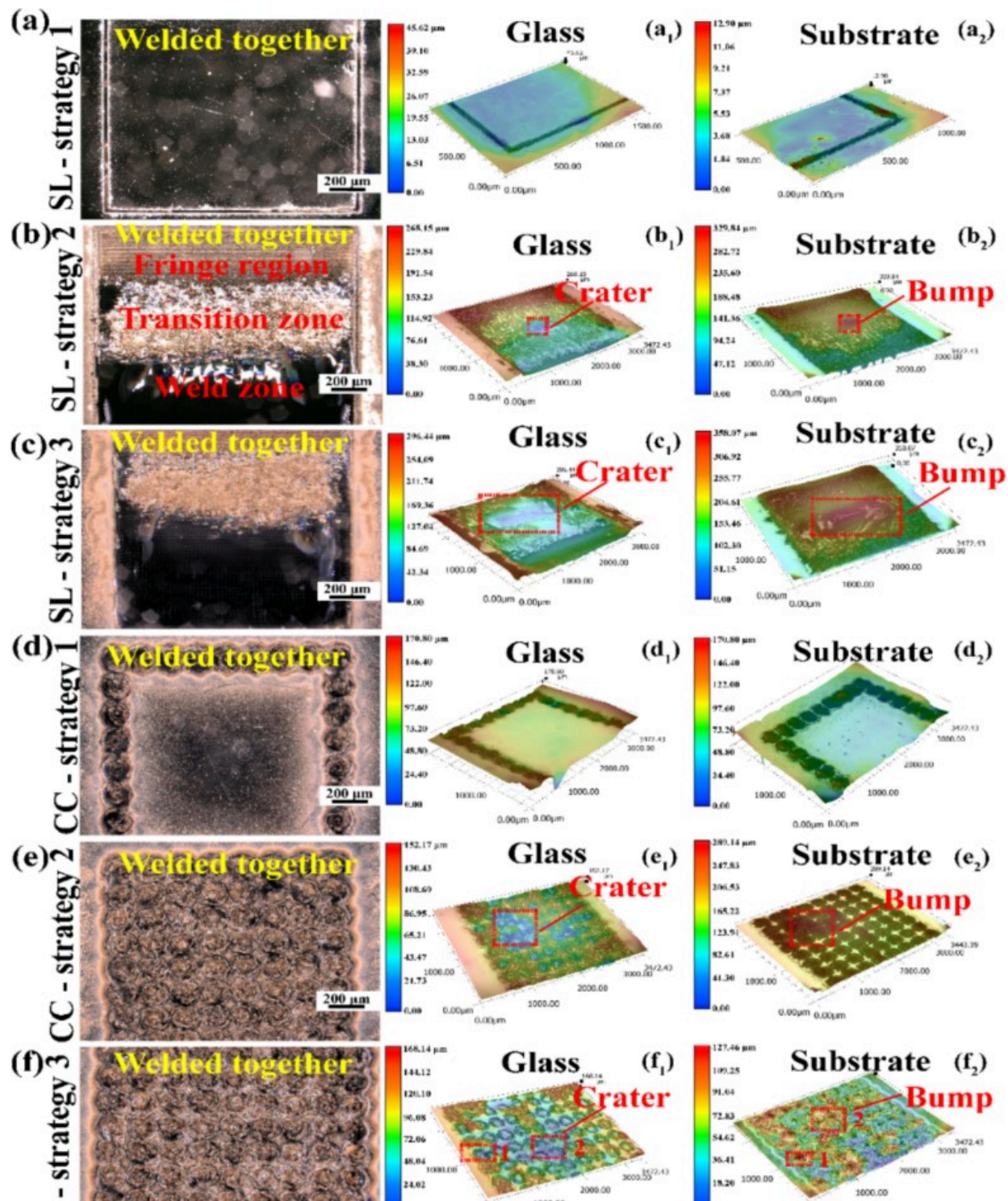
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Fig. 6. (a)–(c) Show the weld joints observed under an optical microscope using CC-strategy 1 with the parameters of Group 1, 3, and 6; (d)–(f) show the results using CC-strategy 2; (g)–(i) show the results using CC-strategy 3. (j) is an enlarged view of the surface holes on the glass corresponding to (c), (f), and (i). (k) illustrates the shear strength measurements for Groups 1–6 using CC-strategy 1–3. The curve represents the overall trend of welding strength. The gray background indicates that Group 4 parameters were selected as the baseline conditions for subsequent welding strategy comparison experiments.

The shear strength results for CC-strategy 1–3 across six parameter groups are presented in Fig. 6(k). Unlike SL-strategy 1 in Fig. 5(k), CC-strategy 1 can generate some shear strength, particularly with Group 3 parameters, where the shear strength for CC-strategy 1 reaches approximately 3.52 MPa. Compared to CC-strategy 1, the shear strength of CC-strategy 2 and CC-strategy 3 is significantly increased across all parameter groups. The overall trend indicates that the shear strength increases initially and then decreases. The maximum shear strength for CC-strategy 3 in Group 5 reaches 14.88 MPa. Although this is slightly lower than the maximum strength observed with SL-strategy 3 in Fig. 5, CC-strategy 3 still offers a significant advantage over CC-strategy 1 and CC-strategy 2. Specifically, CC-strategy 3 enhances shear strength by at least 10% compared to CC-strategy 2, demonstrating its potential to optimize welding performance. Furthermore, CC-strategy 1 presents a valuable option that can achieve welding with less impact on the glass.

As shown in Fig. 5(k) and Fig. 6(k), higher shear strength is generally obtained under the parameters around Group 4. Although Group 5 parameters yielded slightly higher shear strength values compared to Group 4, the differences between the two groups were within the experimental uncertainty range (± 1 MPa). To ensure consistency and minimize additional variables during subsequent experiments comparing different welding strategies, we selected Group 4 parameters as the standardized processing condition. This

choice enables a fairer evaluation of the influence of scanning paths by avoiding compounding effects caused by varying energy input levels. Therefore, in Figs. 5(k) and 6(k), Group 4 is highlighted with a gray background to indicate its use as the baseline parameter set for strategy comparison experiments, rather than representing the group with absolute best performance. Therefore, Group 4 parameters were selected for further analysis of both SL-strategy and CC-strategy. Fig. 7(a)-(c) and Fig. 7(d)-(f) display the 3D morphology of the welds after applying SL-strategy 1–3 and CC-strategy 1–3, respectively, under Group 4 parameters. Fig. 7(a₁)-(f₁) show the 3D morphology of the glass layer after the welded samples were separated, and Fig. 7(a₂)-(f₂) present the 3D morphology of the stainless steel surface. The horizontal and vertical axes represent the width and height of the welded area, while the color scale indicates the height variations across the welded plane. It is evident that the previously flat joining surfaces now appear uneven, indicating successful bonding between the glass and stainless steel and confirming the effectiveness of the applied welding strategies [37].





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Fig. 7. (a)-(f) Are the 2D schematic diagrams of welding using SL-strategy 1–3 and CC-strategy 1–3, respectively; (a₁)-(f₁) are the 3D images of the glass surface after shearing at the interface; (a₂)-(f₂) are the 3D images of the stainless steel surface after shearing.

The crater sizes on the glass surface and the bumps on the stainless steel surface are correlated with glass loss and adherence, as shown in Fig. 7, directly affecting shear strength [38,39]. Comparing SL-strategy 2 and SL-strategy 3, it is evident that after being separated under shear, the stainless steel surface from SL-strategy 2 does not exhibit significant amounts of molten glass adhered to it, unlike SL-strategy 3, as shown in Fig. 7(b₁)(b₂) and Fig. 7(c₁)(c₂). This explains the relatively weaker shear strength of SL-strategy 2. A similar phenomenon is observed in CC-strategy 2 and CC-strategy 3, as shown in Fig. 7(e₁)(e₂) and Fig. 7(f₁)(f₂), where the shear strength of CC-strategy 2 is lower than that of CC-strategy 3. Notably, the glass adhesion to stainless steel in SL-strategy 1 and CC-strategy 1 is much smaller than the other groups, resulting in lower shear strength. However, some glass adhesion can still be observed in CC-strategy 1, confirming that its weld strength is slightly superior than that of SL-strategy 1.

It is observed in SL-strategy 2 and SL-strategy 3 that a layer of glass adheres to the substrate surface at the sheared interface, with a small range of distinct bump structures visible in the central region, as highlighted by the yellow boxes in Fig. 7(b₂) and Fig. 7(c₂). The fused glass is not confined to the focal point area alone. In addition to forming a molten pool of the stainless steel/glass mixture near the laser focus, the glass melted and expanded through heat conduction, subsequently adhering to the substrate surface after shearing [38]. This results in localized regions of pronounced bumps on the substrate surface and corresponding deep craters on the glass. For CC-strategy 2 and CC-strategy 3, a thin layer of glass is primarily observed adhering near the focal point, as highlighted by

the yellow in Fig. 7(e₁₋₂) and Fig. 7(f₁₋₂). Unlike SL-strategy 3, the crater-bump structures in CC-strategy remain highly localized, predominantly concentrated at the center of the concentric circles, and do not exhibit the pronounced results seen in Fig. 7(c). This distinction further substantiates that the overall weld strength of the CC-strategy is lower than that of the SL-strategy.

It can be noticed from Fig. 7(b) and Fig. 7(c) that using SL-strategy 2 and SL-strategy 3 results in a joint structure composed of three distinct layers. Based on the 3D morphology, the lower opaque layer corresponds to the stable welded joint region, where large areas of adhered molten glass are evident on the substrate surface. The middle layer represents the transition zone, where no stable welded joint is formed like a partial bonding area, while a clear laser ablation profile can be observed. In the upper layer, no welded joint is formed in this region. Instead, the laser ablates the surface in a regular straight-line pattern, with the 3D morphology of the interface showing a uniform and even distribution near the focal point. These layers can be categorized as the fringe region (upper layer), the transition zone (middle layer), and the weld zone (lower layer), as shown in Fig. 7(b). Comparing Fig. 7(b) and Fig. 7(c), the transition zone in Fig. 7(b) is noticeably larger, resulting in a smaller melt pool than in Fig. 7(c). This distinction highlights that the width and depth of the weld zone directly influence the shear strength of the joint. Since the primary bonding occurs within the weld zone, the shear strength of the welded joint is positively correlated with the weld zone dimensions. In contrast, the three-layer structure is not observed in Fig. 7(e) and Fig. 7(f), which correspond to the CC-strategy welds. The formation of a well-defined weld zone requires highly concentrated heat transfer, which is inherently challenging for the scanning strategy that applies concentric circles. In CC-strategy 3, only the central part of the concentric circles achieves sufficient heat concentration, resulting in a small localized crater-bump structures at the center. This observation further confirms that CC-strategy welds exhibit significantly lower shear strength than SL-strategy welds due to reduced heat concentration and limited weld zone formation.

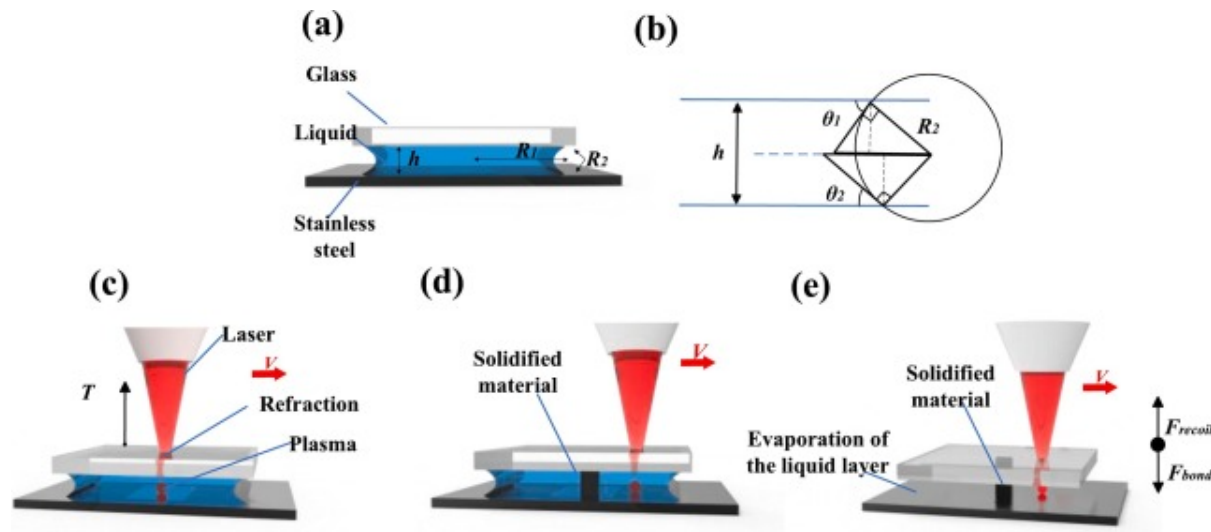
With Strategy 2, excessive energy input leads to significant heat accumulation, resulting in fluid evaporation and excessive reverse thrust within the molten pool. These factors contribute to weld failure at the initial position. In contrast, Strategy 3 effectively mitigates this issue by utilizing the outer frame area to control the welding process. Although the gap fluid eventually evaporates, Strategy 3 maintains optical contact at the outer frame via the initial joint formed. This initial joint stabilizes the welding process, and the subsequent filling enlarges the molten pool area, ensuring a robust connection between the samples [40]. As a result, the shear strength and stability of the welded joints are significantly enhanced with the method using Strategy 3.

In summary, upon a comparison of the straight-line welding element (SL-strategy) and the concentric circle welding element (CC-strategy), the overall welding strengths are ranked from smallest to largest as follows: SL-strategy 1, CC-strategy 1, CC-strategy 2, CC-strategy 3, SL-strategy 2, and SL-strategy 3. The SL-strategy demonstrates a higher upper limit of shear strength than the CC-strategy, with a maximum shear strength of 19.73 MPa. This observation highlights the effectiveness of LFTW and validates the method using the straight line as the fundamental welding element. Among the various strategies, Strategy 3 stands out as it significantly improves shear strength compared to Strategies 1 and 2, thereby underscoring its superiority. Although the CC-strategy series do not achieve the exact upper limit of weld strength as the SL-strategy series, they exhibit a relatively high lower limit of shear strength, providing the advantage of stable welding over a small area of influence. Strategy 3 exhibited the best welding performance in both concentric circle and straight-line configurations. The enhancement in strength and morphological stability was particularly notable under optical contact conditions. These results suggest that a sequential scan of the outer frame followed by internal filling can significantly improve both weld strength and process stability compared to conventional single-pattern strategies. Based on these findings, the study intends to continue using SL-strategy 3 while further exploring the impact of pressurization under varying operating conditions to improve the joint performance even further.

The principle underlying optical contact formation via the LFTW method is based on capillary forces provided by the anhydrous ethanol liquid film. A thin liquid film of anhydrous ethanol is introduced between the glass and the stainless steel plate. Due to the excellent wetting properties of anhydrous ethanol with both glass and stainless steel, the liquid film forms a complete contact interface with the glass under sufficient applied pressure, thereby eliminating any air layer, as illustrated schematically in Fig. 8(a). The liquid film acts as a liquid bridge between the plates. Based on the previous studies [32,41], the surface tension at the gas-liquid interface during the formation of the liquid bridge can be directly described using the Young-Laplace equation, expressed as Eq. (3) [42].

$$\Delta p = \sigma \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad (3)$$

where Δp is the pressure difference across the interface, σ is the surface tension of ethanol, and R_1, R_2 are the principal radii of curvature of the liquid surface.



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Fig. 8. (a) Force diagram and system schematic of the upper glass layer assisted by a liquid film; (b) radius diagram of the concave region at the liquid-gas interface of the liquid film; (c) schematic of recoil pressure after the initiation of welding; (d) schematic of the joint formed after welding; and (e) schematic of the weld following the evaporation of the liquid film.

Given that the lateral radius R_1 is much larger than the vertical radius R_2 in our configuration (thin film geometry), the reciprocal of R_1 becomes negligible in comparison to the reciprocal of R_2 , thereby simplifying Eq. (3) to Eq. (4) [43]. The contact angles θ_1 and θ_2 represent the angles at which the ethanol contacts the upper glass layer and the lower stainless steel layer respectively, influencing the value of R_2 . Since the ethanol solution wets both materials effectively, the liquid bridge forms a depressed internal surface, with its edge morphology schematically illustrated in Fig. 8(b).

$$\Delta p = \sigma \left(\frac{1}{R_2} \right) = \frac{\sigma(\cos \theta_1 + \cos \theta_2)}{h} \quad (4)$$

where h is the thickness of the ethanol film.

This positive capillary pressure effectively compresses the glass against the stainless steel substrate, compensating for the absence of external mechanical clamping and maintaining optical contact throughout the welding process.

When a nanosecond laser heats the liquid film, it induces a localized temperature gradient, resulting in temperature-driven convection currents within the liquid and vapor outburst recoil, as illustrated in Fig. 8(c). Under laser heating, the material expands, causing the gap to widen. However, when the laser reaches the interface between the glass and stainless steel, it is absorbed linearly by the metal through free electrons, leading to avalanche ionization and the eventual formation of a dense plasma state. The energy and significant thermal accumulation from the multi-pulse laser create a larger melt pool. The molten material and plasma are ejected to fill the voids between the two materials, forming a joint [40]. The solidified joint bonds tightly, providing tension that

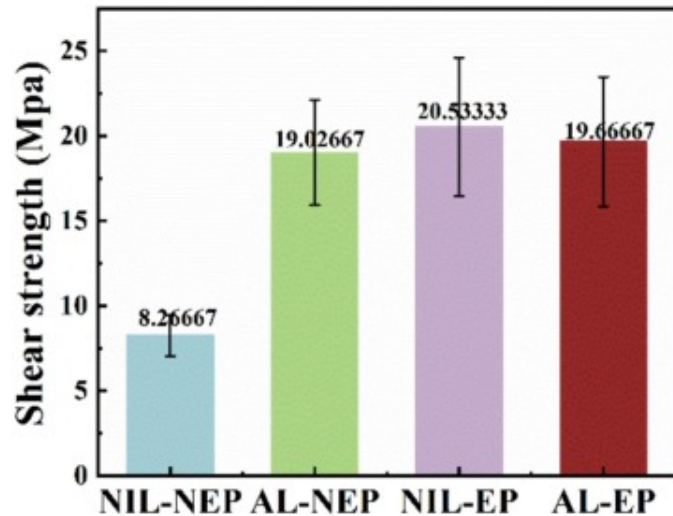
counteracts the recoil pressure T , thus maintaining optical contact. This process facilitates the subsequent welding steps, as depicted in Fig. 8(c) and Fig. 8(d). Subsequently, upon laser irradiation, the anhydrous ethanol film was rapidly heated and vaporized due to its low boiling point, as illustrated in Fig. 8(e). The evaporation process generates transient vapor pressure at the glass-metal interface. This vapor pressure induces an outward recoil force (F_{recoil}) that tends to separate the glass from the stainless steel, potentially compromising optical contact and weld stability. However, the initial optical bonding at the outer frame effectively locks the boundary and constrains the vapor expansion, maintaining intimate contact across the interface.

During the welding process, the ethanol liquid film evaporates rapidly upon initial laser irradiation, especially along the outer frame, which is scanned first. This leads to early interface bonding at the edges and helps maintain optical contact during the interior scanning stage. Since the interior is scanned repeatedly (50–90 passes), any residual ethanol is removed quickly after the first scan. Thus, no intermediate liquid or residual layer remains between the glass and stainless steel after welding. The final interfacial thickness is determined by the weld seam itself and remains relatively uniform, independent of the initial variation in the liquid film thickness.

3.2. Evaluation of joint quality under different operating conditions

Given the cumulative expansion of thermal effects during welding and the eventual degradation of the liquid layer due to evaporation over time, external pressurization was implemented in this study to further minimize the gap and maintain continuous optical contact. Fig. 9 illustrates the variations in shear strength under four different operating conditions after adjusting the operating parameters. The shear strength in the NIL-NEP condition is markedly lower, approximately half that of the other three conditions. While the differences in shear strength among AL-NEP, NIL-EP, and AL-EP are relatively minor, certain distinctions remain notable. Specifically, the AL-NEP condition achieves a shear strength of 19.02 MPa. In comparison, the NIL-EP condition, which applies external pressure without an intermediate layer, results in a slightly higher shear strength of

20.53MPa. Meanwhile, the AL-EP condition, which incorporates an alcohol-based intermediate layer along with external pressure, reduces the shear strength to 19.66MPa. Despite this reduction, the AL-EP condition still exhibits a marginal improvement over the AL-NEP condition.

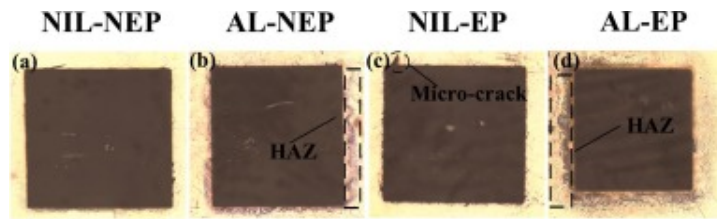


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Fig. 9. Schematic diagram of the variation of shear strength under four operating conditions.

Fig. 10(a)-(d) display the 2D surface morphology of the welded samples under the four conditions. Both Fig. 10(b) and Fig. 10(d) reveal a larger HAZ compared to the other two methods. This increased HAZ is attributed to the carbonization of the alcohol layer, which leads to greater thermal diffusion. In contrast, Fig. 10(a) and Fig. 10(c) show visible micro-cracks along the edges, attributed to the elevated energy input during the welding process. The higher thermal stresses and localized stress concentrations at the corners of the laser-scanned area ultimately leading to crack formation.



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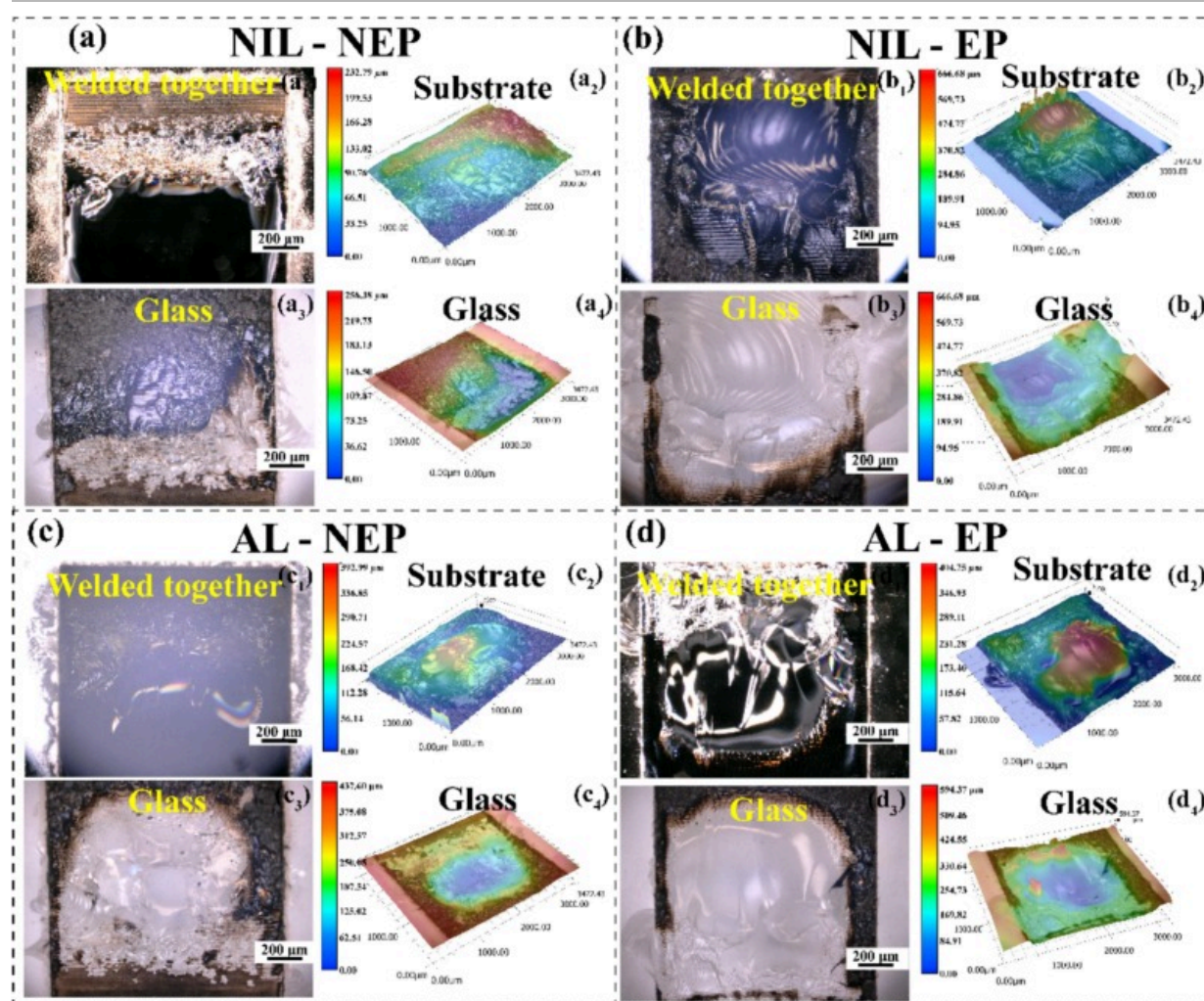
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Fig. 10. (a-d) Are the 2D images of the samples welded together under the four operating conditions.

Due to the limitations in the temporal and spatial resolution of conventional thermal imaging equipment (e.g., MAG32 camera), accurate real-time temperature measurement during nanosecond pulsed laser welding was not feasible in this study. Consequently, thermal field analysis based on direct measurement was omitted. Future work will focus on applying high-speed infrared thermography and time-resolved diagnostics to better capture the transient thermal behavior.

Fig. 11(a)-(d) presents the 2D profiles of the welded joints (a_1 , b_1 , c_1 , d_1) and glass sheets after separation (a_3 , b_3 , c_3 , d_3), along with the 3D morphology of the stainless steel substrates (a_2 , b_2 , c_2 , d_2) and glass surfaces (a_4 , b_4 , c_4 , d_4) under the four operating conditions. In Fig. 11 (a_1)(b_1)(c_1)(d_1), the central solid black zone represents the tightly welded joint, corresponding to the weld zone as described in Section 3.1, where a mixed molten pool is formed. This molten pool obstructs light transmission through the welded zone, preventing visibility of the laser writing marks on the underside. Around the unwelded areas of the glass in Fig. 11(a_3)(b_3)(c_3)(d_3), carbonized traces originated from thermal melting are observed, which indicates the fringe region. Post-experimental observations revealed localized dust discoloration in these areas, suggesting that certain joints formed by dust remelting during the ablation process [18]. The residual carbonized areas are attributed to the heat accumulation and the intense, unstable impact of plasma.

These effects hinder the formation of a stable joint, causing the glass to adhere unevenly to the stainless steel surface rather than forming a uniform weld.



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Fig. 11. (a)-(d) Are the schematic diagrams of the welds under the four operating conditions. (a₁), (b₁), (c₁), and (d₁) are the 2D images of the welded joints. (a₂), (b₂), (c₂), and (d₂) are the 3D images of the fracture surfaces on the stainless steel after shearing

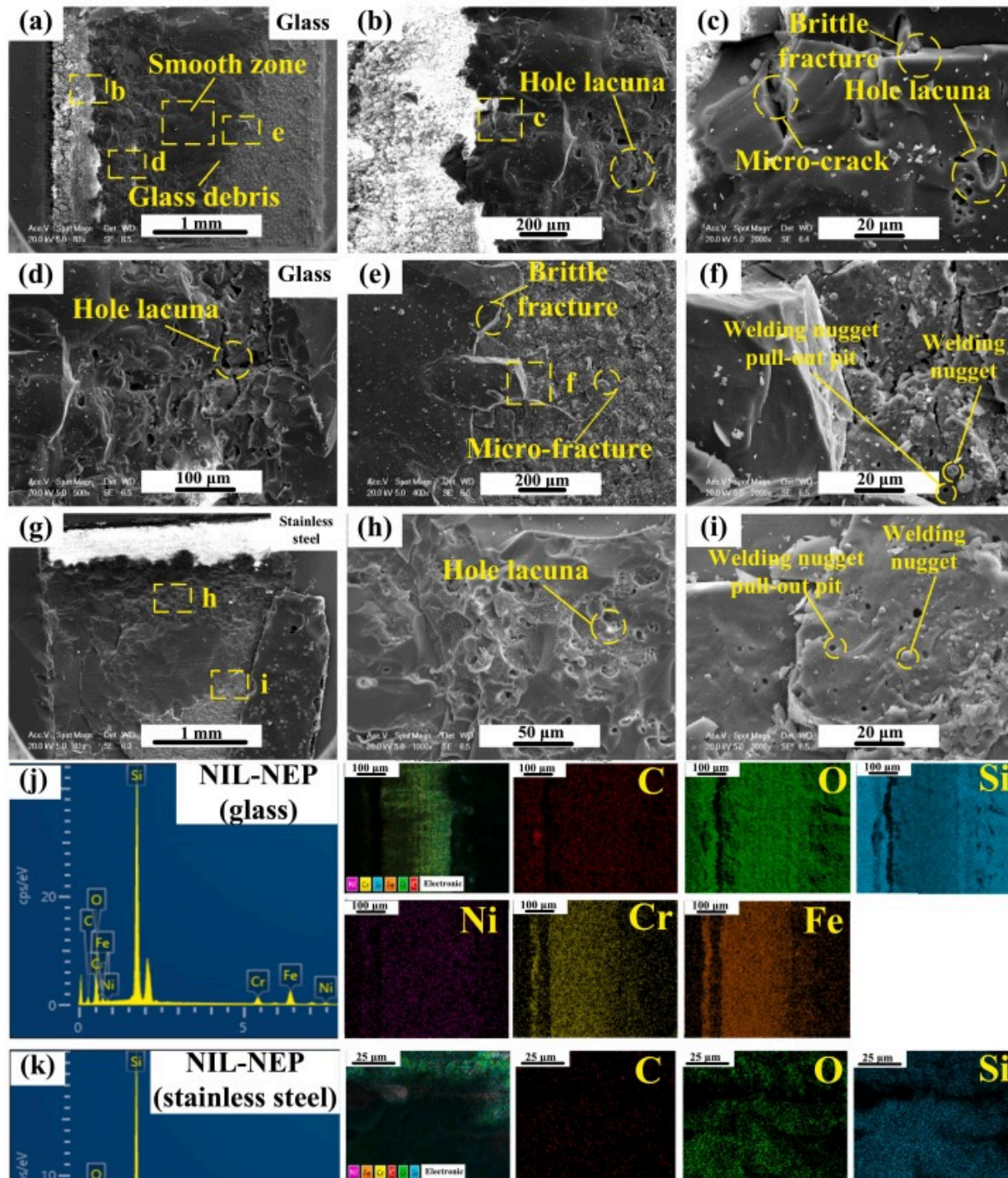
separation. (a_3) , (b_3) , (c_3) , and (d_3) are the 2D images of the fracture surfaces on the glass after shearing separation. (a_4) , (b_4) , (c_4) , and (d_4) are the 3D images of the fracture surfaces on the glass after shearing separation.

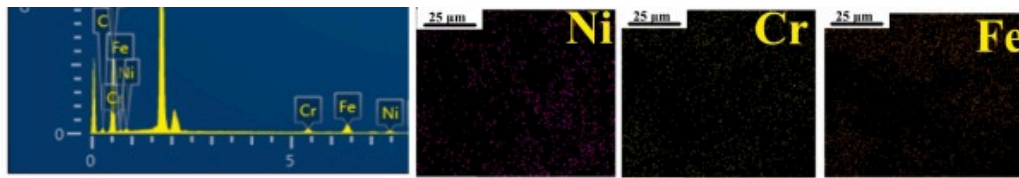
Similar to [Section 3.1](#), the shear strength can be inferred from the size of the craters on the glass surface and the prominence of bumps on the stainless steel substrate, as depicted in the 3D diagrams in [Fig. 11](#). Under the NIL-EP, AL-NEP, and AL-EP conditions, the 3D diagrams distinctly exhibit large central craters on the glass surface and pronounced bumps extending beyond the focal point on the substrate surface. These morphological features are indicative of welded joints with substantially higher shear strength. In contrast, under the NIL-NEP condition, the welded zone on the substrate surface is confined to the immediate vicinity of the focal point, with only a tiny portion of the glass adhering to the substrate. This result highlights the notably lower weld strength achieved under the NIL-NEP condition compared to the other three methods.

After measuring the shear strength of the joints, the fracture profiles and morphologies were analyzed under various operating conditions. [Fig. 12](#) presents SEM images of the weld interface morphology under the NIL-NEP condition. [Fig. 12\(a\)-\(f\)](#) and [Fig. 12\(g\)-\(i\)](#) depict the fracture morphology on the glass side and stainless steel side, respectively. In [Fig. 12\(a\)](#), a smooth zone is visible at the center, resulting from the crater structure formed by internal tearing of the glass. Due to its internal origin of glass, this zone appears smoother than the outer edges. Similarly, the smooth zone in the central region of [Fig. 12\(g\)](#) corresponds to the portion of glass that adhered to the stainless steel. Adjacent to the smooth zone, a substantial amount of glass debris is present, which suggests a weak weld strength. [Fig. 12\(b\)](#), [\(d\)](#), and [\(e\)](#) provide the magnified views of specific regions in [Fig. 12\(a\)](#). In [Fig. 12\(b\)](#), two distinct areas are evident. The left region corresponds to the fringe region (as discussed in [Section 3.1](#)), where excessive heat accumulation scorches the welded area, causing material spattering and preventing the formation of a stable welded joint. The EDS scan of the fringe region is presented in [Fig. 12\(j\)](#), confirming the distribution of Ni, Cr, and Fe elements from the stainless steel, indicating the stainless steel elements have intermixed with glass material in the splash.

The right region in Fig. 12(b) exhibits numerous brittle fractures and irregular pores, which are characteristic of hole lacunas. A closer view in Fig. 12(c) reveals clear evidence of melting and subsequent solidification. The irregular hole lacunas in this area are attributed to laser ablation during welding that generates upward vapor and laser-induced cavitation bubbles at the seam. The rapid cooling of the molten pool following laser irradiation prevents complete backfilling of these pores, leading to their entrapment within the weld [44]. The formation of the fracture is primarily due to the accumulation of laser energy input on the glass side, resulting in defects such as micro-cracks during cooling. These structural inconsistencies significantly degrade the shear strength of the weld and hinder the formation of a dense joint. Fig. 12(d) highlights the transition zone (as previously discussed in Section 3.1), where numerous irregular hole lacunas can also be observed, consistent with the corresponding transition zone in Fig. 12(h). Fig. 12(e) depicts the transition zone on the other side, which contains brittle fractures adjacent to the smooth zone at the center. Fig. 12(f), a closer inspection of Fig. 12(e), reveals numerous rounded and regular pits and bumps (approximately 3–4 μm in size) formed along the weld joint during separation. These features are welding nuggets and welding pull-out pits that form at the seam near the focal point [45], also evident in Fig. 12(i), (k). The EDS results in Fig. 12(k) demonstrate a uniform distribution of metal elements within the welding nuggets, confirming a typical welded joint where the glass adheres to the stainless steel [46]. Additionally, glass debris and microfractures are visible, indicating incomplete welding nugget formation and reduced weld strength due to these defects. In Fig. 12(i), the corresponding stainless steel surface displays regular welding nuggets and pull-out pits, contributing to a modest weld strength. However, based on the 3D morphology in Fig. 11, only the smooth zone exhibits a relatively strong welded joint. Under NIL-NEP conditions, this smooth zone is confined to a small central region, while the surrounding areas contain regions that compromise weld integrity. These weak regions are prone to fracture during shear strength testing, resulting in reduced overall weld strength.

NIL-NEP





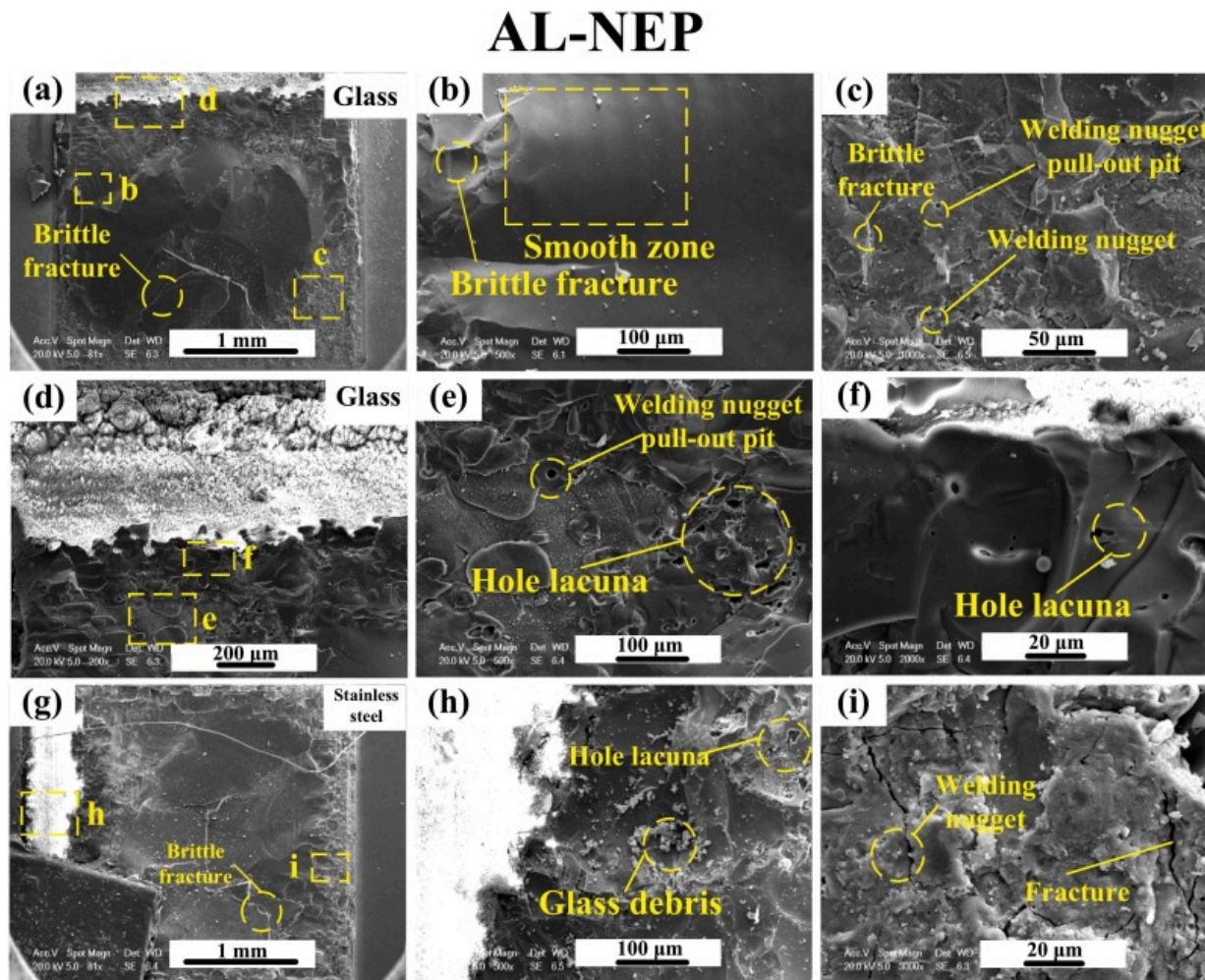
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Fig. 12. Under the NIL-NEP condition: (a)-(f) are the SEM images of the glass side, (g)-(i) are the SEM images of the stainless steel side, and (j) is an elemental distribution map obtained by EDS mapping of the left region in (a), and (k) is an EDS scan image of the fringe region on the stainless steel.

The SEM images of the welded joint morphology under AL-NEP conditions are shown in Fig. 13. Fig. 13(a)-(f) and Fig. 13(g)-(i) illustrate the fracture morphology on the glass and stainless steel sides, respectively. The region outlined by the yellow box in Fig. 13(a) is further detailed in Fig. 13(b), (c), and (d). Additionally, Fig. 13(e) and (f) provide an enlarged view of the area highlighted in Fig. 13(d). In Fig. 13(a), an extensive smooth zone can be observed at the center of the weld joint. Compared to Fig. 12, the amount of glass debris surrounding this zone is significantly reduced, which suggests that the overall weld strength under the AL-NEP condition is higher than that under the NIL-NEP condition. However, certain defects persist. In Fig. 13(b), the detailed smooth zone, similar to that observed in the center of Fig. 12(a), is characterized by brittle fracture caused by crack propagation along the glass interior. Fig. 13(c) highlights the welding nugget and pull-out pit formed near the weld focal point, which is a typical feature of the welded joint interface. The weld strength in this area is notably low, attributed to the accumulation of char and debris, corresponding to the right region in Fig. 12(a). Meanwhile, Fig. 13(d) reveals a distinct division between the upper and lower layers: the upper part corresponds to the fringe region, while the lower part represents the transition zone. This area exhibits numerous brittle fractures and irregular holes, preventing the formation of a structurally sound welded joint. The enlarged views in Fig.

13(e) and (f) emphasize brittle fractures and irregular hole lacunas of the transition region. These imperfections, clearly visible in Fig. 13(e) and (f), negatively impact shear strength and hinder the formation of a dense welded joint. On the stainless steel side, Fig. 13(g) presents the overall fracture morphology, with Fig. 13(h) and (i) offering enlarged views of specific regions. Fig. 13(h) corresponds to the area as depicted in Fig. 13(d), displaying the fringe region on the left and the transition zone on the right. Numerous hole lacunas and glass debris, approximately 20 μ m in size, are observed in the transition zone, further obstructing the formation of a robust welded joint. Fig. 13(i) highlights the welding nuggets on the stainless steel surface, while the presence of extensive fractures limits the weld strength and restricts the expansion of the welded joint. Most regions near the focal point, including the fringe region and transition zone, represent the early stages of welded joint formation and contribute minimally to weld strength. Although defect areas remain, they have been considerably reduced in comparison to the NIL-NEP condition, and the smooth zone has increased. This indirectly indicates an improvement in weld strength under AL-NEP conditions.



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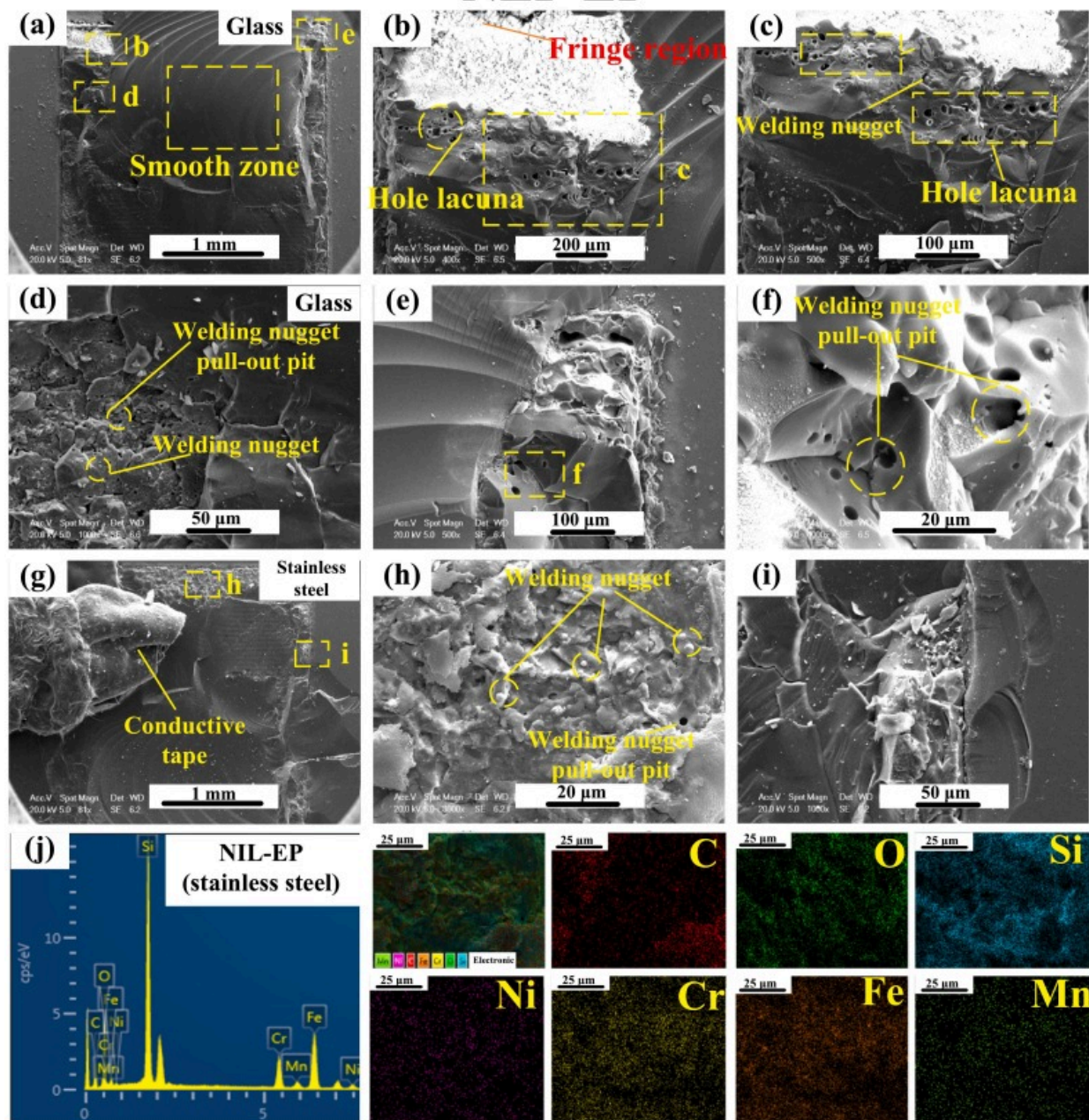
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Fig. 13. Under the AL-NEP condition: (a)-(f) are the SEM images of the glass side, and (g)-(i) are the SEM images of the stainless steel side.

The SEM images of the morphology at the welded joints under NLE-EP conditions are shown in Fig. 14. Fig. 14(a) displays larger smooth zones, which slightly exceed those observed under AL-NEP conditions. The defective areas are limited to the four corners of

the stainless steel, suggesting a marginally higher weld strength. The detailed analysis of these defective areas is as follows. Fig. 14(b), (d), and (e) provide magnified views of Fig. 14(a). Fig. 14(b) highlights the fringe region and transition zone, while Fig. 14(c) presents an enlarged view of the transition zone in Fig. 14(b), where numerous distinct hole lacunas approximately 20 μm in size are visible. These hole lacunas are formed during the melting and re-solidification of the welded joint material. The transition zone is surrounded by a smooth inner glass layer, separated by cracks along the brittle fracture. Fig. 14(d) resembles the large area on the right side of Fig. 12(e), exhibiting microcracks and glass debris. Small-scale spatter particles from the welding process [47] are also visible. These spatter particles and debris contribute to weld instability and reduce weld strength [48,49]. This part of the welded joint represents the early stage of joint formation, where the molten pool remains unstable on both the glass and stainless steel sides. The presence of brittle phases further decreases the weld strength. Fig. 14(e) and 14(f) focus on the center of a stable welded joint. Fig. 14(f) clearly shows regular welding pull-out pits, which is considered as the characteristic of a more stable joint formation. As shown in Fig. 14(g), most of the smooth zone in the center consists of a substantial amount of glass adhered to the stainless steel. During fracture, the joint exhibits brittle behavior, and the fracture propagates along the inside of the glass. Laser scanning marks remain visible beneath the glass after the fracture. At this stage, the shear strength primarily depends on the interaction between the expanded molten pool inside the glass and the glass surface. In Fig. 14(h), the micro-bump represents the welding nugget. The EDS surface scan in Fig. 14(j) confirms a homogeneous distribution of elements throughout the region, with a uniform mixture of Si and metallic elements in the stainless steel, characteristic of the region after weld pull-out. Fig. 14(i) shows the fracture edge of the stainless steel surface adhering to the glass at the focal point.

NLP-EP

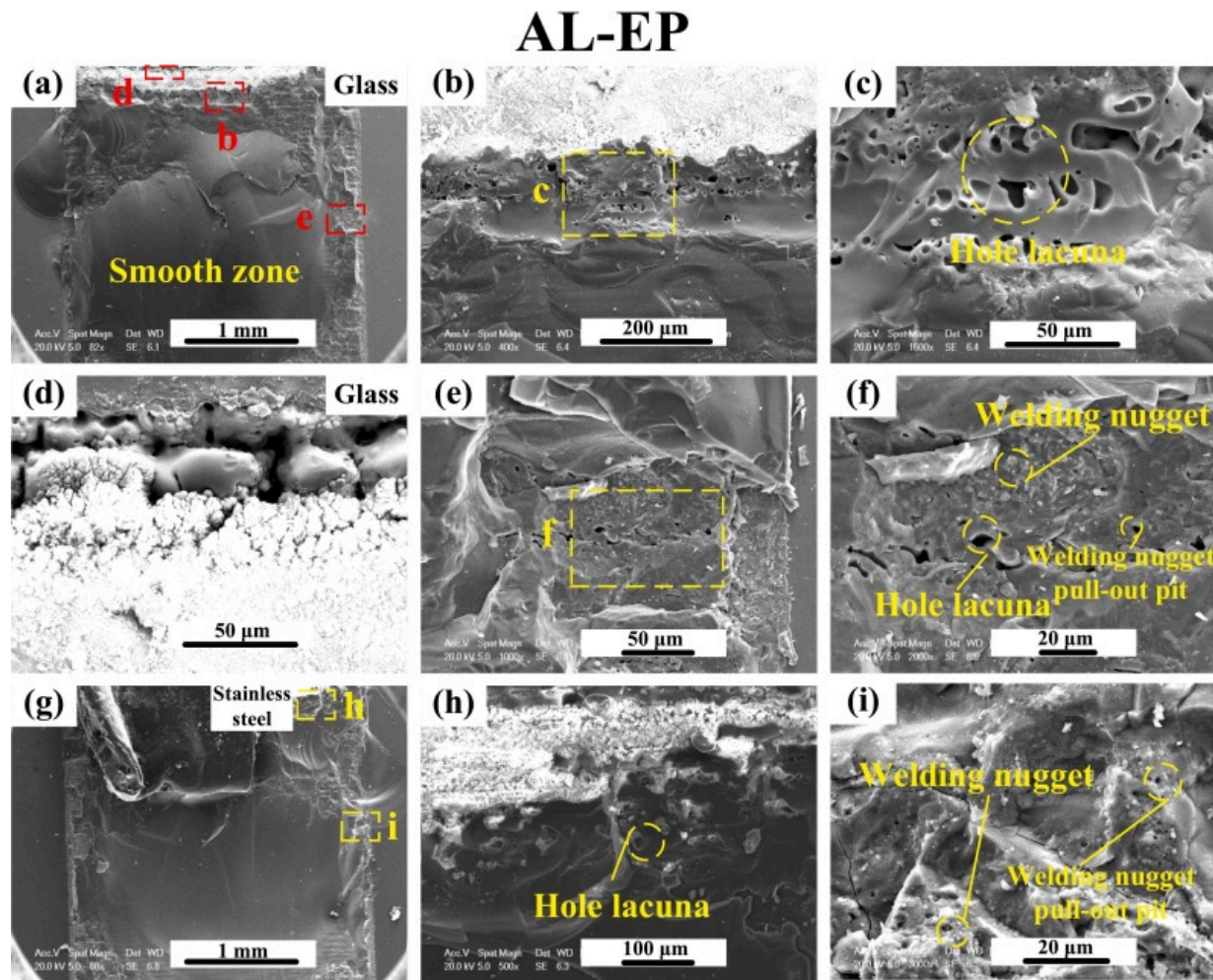


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Fig. 14. Under the NIL-EP condition: (a)-(f) are the SEM images of the glass side, (g)-(i) are the SEM images of the stainless steel side, and (j) is an elemental distribution map obtained by EDS mapping of the left region in (h).

The SEM image of the morphology at the welded joint under the AL-EP condition is shown in Fig. 15. Compared to NIL-EP, Fig. 15(a) reveals that the smooth zone is smaller than that shown in Fig. 14(a), and the surrounding fringe region and transition zone exhibit more fractures. As a result, the weld strength under AL-EP is comparable but slightly lower than that under NIL-EP. A detailed analysis of the defective areas can be found as follows. Fig. 15(b) and (c) provide an enlarged view of the glass transition region, while Fig. 15(h) shows the corresponding region on the stainless steel. The fracture appears relatively flat in Fig. 15(c) and (h), indicating that this part forms the welded joint connecting the stainless steel. Many hole lacunas of about 20µm can be observed, surrounded by some welding nugget pull-out pits. Fig. 15(d) shows the fringe region where a deep crater has formed due to the repeated laser ablation in the same area. Fig. 15(e) and (f) depict the separation interface from the welded joint, which is adjacent to the smooth zone. Similar to Fig. 14(e) and (f), this region also features the welded joint, with visible welding nuggets and welding nugget pull-out pits. This corresponds to the phenomenon observed in the stainless steel region as shown in Fig. 15(i). The slightly lower weld strength compared to NIL-EP may be attributed to the addition of alcohol during the welding process. The interaction between the laser plasma and the liquid alcohol layer generates more cavitation bubbles, leading to the formation of additional hole lacunae.



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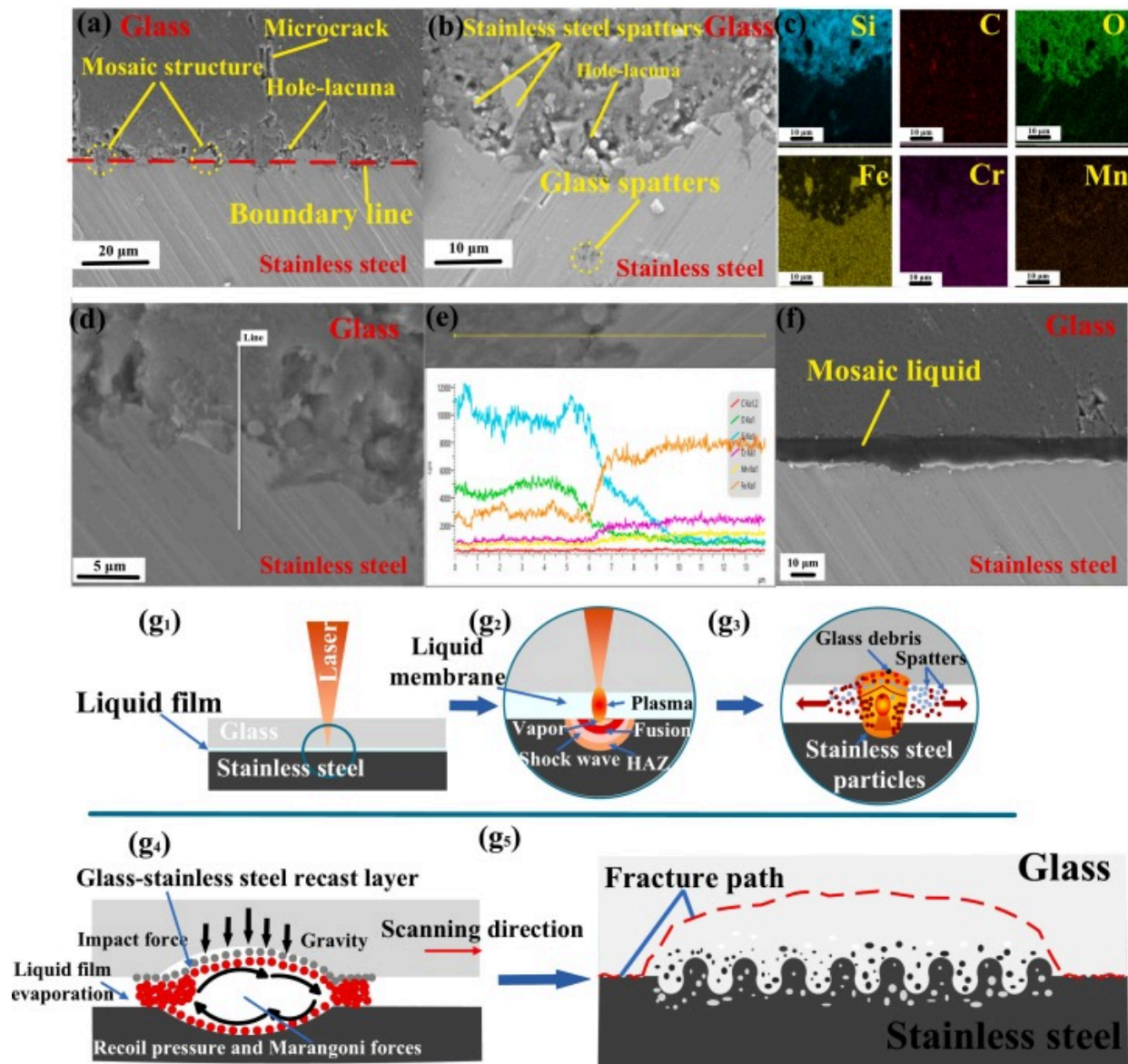
Fig. 15. Under the AL-EP condition: (a)-(f) are the SEM images of the glass side, (g)-(i) are the SEM images of the stainless steel side, and (j) is an elemental distribution map obtained by EDS mapping of the left region in (h).

In summary, a large number of smooth regions are formed under AL-NEP, NIL-EP, and AL-EP conditions, in contrast to the NIL-NEP condition. This observation aligns with the

welding strength trends depicted in [Fig. 9](#). Despite the presence of defect areas, analyzing the formation process of welded joints provides valuable insight into optimizing laser welding parameters.

3.3. Mechanism analysis of laser welding interface

The fundamental principle behind nanosecond laser or long-pulsed laser welding of transparent materials and metals involves the precise focusing of the laser beam at the bond interface. This process depends on the action mechanisms of melting and re-solidification, as well as the laser absorption mechanism at the interface. Specifically, the laser absorption is driven by linear absorption and collisional ionization within the material [50]. Based on this laser absorption mechanism, the schematic diagram of the nanosecond laser welded joint between glass and stainless steel under the AL-NEP condition (using Group 4 parameters) is shown in [Fig. 16](#).



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Fig. 16. (a), (b), (d), (f) Depict the welding interface profiles obtained through SEM; (c), (e) present EDS scanning results corresponding to (b) and (d) respectively; (a)-(f) are

obtained using the parameters of Group 4 under AL-NEP condition. (g₁)-(g₅) show the schematic diagram of welding.

Fig. 16(a) presents a SEM micrograph of the welded joint section, which reveals a periodic mosaic structure at the glass and stainless steel interface. Additionally, micro-lacunae and microcracks are observed in the upper part of the glass. Fig. 16(b) provides a magnified view of a single mosaic structure from Fig. 16(a), with the corresponding EDS surface scanning results are shown in Fig. 16(c). These results indicate that the mosaic structure forms due to the cooling process of the molten pool at the interface between the stainless steel and glass. The EDS analysis further confirms that stainless steel and glass particles have mixed on opposite sides of the interface. Line EDS scans of the mosaic structure, as shown in Fig. 16(d) and Fig. 16(e), reveal an increase in metallic elements at the joint, while glass elements gradually decrease until stabilization. This intermingling of metal and glass elements indicates the formation of a stable welded joint. Fig. 16(f) shows a region where the welding was unsuccessful, marked by a gap in the middle. During sample embedding, this gap was filled with an inlay solution. In this region, the glass and stainless steel interface lacks the mosaic structure observed in successful welds. Instead, only the upper surface of the stainless steel is partially indented by the laser, while the glass remains relatively smooth.

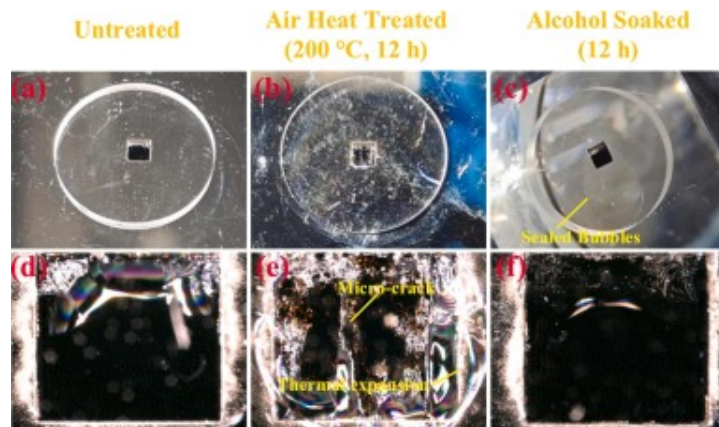
The schematic of the welding process is shown in Fig. 16(g₁-g₅). Initially, the nanosecond laser energy is absorbed by the surface of the stainless steel, penetrating a few hundred nanometers to a few micrometers. This causes plasma formation and the development of a small initial molten pool are shown in Fig. 16(g₁₋₃). The behavior of the plasma plays a critical role in weld formation. It aids material mixing and the creation of strong joints but also generates significant spatters from glass debris and stainless steel particles. Moreover, the intense plasma activity, coupled with optical contact conditions and limited gaps, induces disordered plasma behavior and excessive heat accumulation. This results in the emission of recoil vapor, which leads to melt pool sputtering and shock waves, forming HAZ in the surrounding area [51,52].

Fig. 16(g₄) illustrates the schematic of melt pool formation at the interface. Following the initial laser scan, the nanosecond laser does not directly create a welded joint. Instead, a recast layer forms on the lower surface of the glass through a melting and solidification process. In the subsequent scans, both the glass and the recast layer are re-melted and splashed onto the stainless steel surface due to the impact of the nanosecond laser and its own gravity, accompanied by the formation of some hole lacunas. Additionally, the glass in the laser-modified area loses its transparency and begins to absorb more laser energy. In the regions where laser spots overlap, increased molten materials are produced and aggregates. Under the influence of recoil pressure and Marangoni forces, the molten material flows into gaps around areas with insufficient scanning, leading to re-solidification and formation of small welding nuggets, which gradually grow. This process ultimately forms a mosaic structure weld [53], as shown in Fig. 16(g₅). A fully developed mosaic structure is observed in the center part of the joint, and the weld strength primarily relies on this region. During the shear testing, the joint fractured from the inside of the upper layer of glass, leaving a smooth zone. In contrast, the weld strength at the two sides of the joint was lower due to incomplete mosaic formation. The fracture propagated along the interface, creating the fringe regions and transition zones. The primary fracture occurred in the brittle phase, with only small portions of the melt pool, welding nugget, and pull-out pit visible.

In summary, when the strength of the weld is considerably lower than that of the substrate materials, the fracture occurs at the interface, resulting in the formation of a fringe region and transition zone on the separated surfaces. In contrast, when the weld strength significantly exceeds that of the substrate materials, the fracture propagates within the substrate materials, following a path along its interior, and resulting in a smooth zone on the separated surface. Under the AL-NEP condition, both fracture modes can coexist, with the relative proportions of each mode influencing the overall weld strength.

3.4. Durability evaluation of the weld joint

To evaluate the durability of the AL-NEP welded joints, two accelerated environmental strength tests were conducted: (1) heat treatment in air at 200 °C for 2–12 h, and (2) immersion in ethanol at room temperature for up to 12 h. The surface condition and morphology were analyzed by digital stereomicroscope (VHX-6000, Keyence) as shown in Fig. 17.



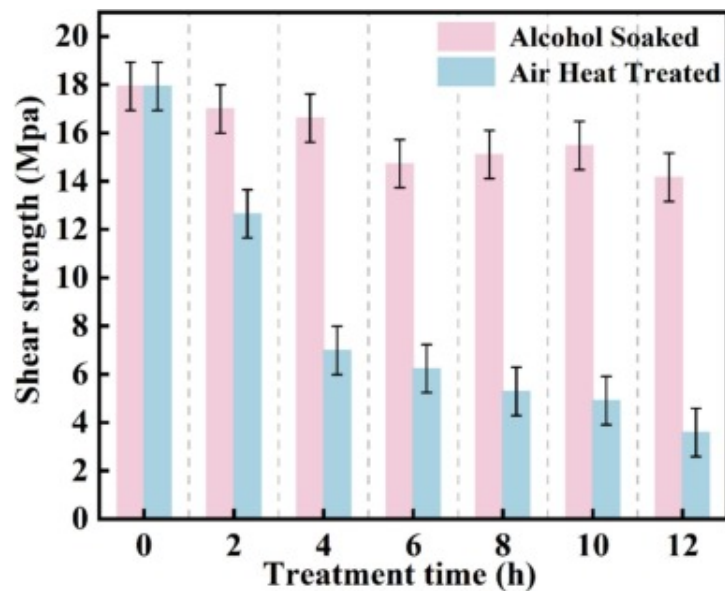
Download: [Download high-res image \(327KB\)](#)

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Fig. 17. Surface morphology of AL-NEP joints under three different post-treatment conditions: untreated, air heat treated (200 °C, 12 h), and ethanol-soaked (12 h). Images (a)–(c) were captured using a smartphone camera, while (d)–(f) were obtained via digital stereomicroscopy. Specifically, (a)(d) correspond to untreated samples, (b)(e) to air heat-treated samples, and (c)(f) to ethanol-soaked samples.

Fig. 17(a)(d) illustrate the as-welded sample, with a clearly dense and continuous weld zone. After 12 h of thermal exposure, Fig. 17(b)(e) reveal the formation of micro-cracks and thermal expansion damage around the glass periphery, indicating that excessive heating degraded the weld zone. In contrast, samples soaked in ethanol (Fig. 17(c)(f)) retained sealed microbubbles and a dense weld interface, with no signs of delamination or chemical infiltration.

The corresponding shear strength results are summarized in Fig. 18. In the heat-treated group, the shear strength dropped rapidly to about 13 MPa after 2h and continuously decreased to about 4 MPa by 12h, indicating a strength reduction of about 77%. However, in the ethanol-soaked group, the shear strength remained relatively stable, with only a decrease of about 16% after 12h compared to the untreated condition. These results confirm that the weld zone is chemically resistant and retains mechanical integrity in alcohol environments, while high-temperature heat treatment significantly weakens the interfacial bond.



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Fig. 18. Shear strength variation of AL-NEP joints under air heat treatment and alcohol soaking over time (0–12h).

This demonstrates that the AL-NEP joints exhibit excellent chemical stability and acceptable thermal resistance, supporting their potential use in chemically harsh or moderately elevated temperature environments.

4. Conclusion

In this study, a novel nanosecond laser welding technique based on LFTW is proposed, for efficiently joining glass to 304 stainless steel plates. Low-cost and fixture-free optical contact conditions were achieved by replacing conventional external pressure with the surface tension and capillary action of the liquid film. Two welding element configurations (straight lines and concentric circles) and three scanning strategies were systematically analyzed through experiments. The experimental results demonstrate that the scanning strategy (SL-strategy 3) using a straight-line element combined with an external frame and internal filler significantly enhances the weld strength, achieving a maximum shear strength of 19.73 MPa. Additionally, by comparing four working conditions, without an intermediate layer versus a liquid film of alcohol, and with or without external pressure, it was found that the LFTW method provides a weld strength comparable to that of the externally pressurized condition without external pressure. However, when external pressure was applied in combination with the liquid film, the weld strength slightly decreased. This LFTW technique presents substantial potential for industrial applications and offers a novel solution for the welding of dissimilar materials. In addition, durability evaluations under thermal and chemical stress conditions confirmed the reliability and stability of the welded joints, further supporting the practical feasibility of the LFTW technique. The main conclusions are as follows:

1. A comparative analysis of two welding strategies, straight line (SL-strategy) and concentric circle (CC-strategy), reveals that the combined scanning strategy of the straight line element with “outer frame scanning and inner filling” (SL-strategy 3) significantly enhances weld performance, achieving a maximum shear strength of 19.73 MPa, which is at least 40% higher than the other strategy. Additionally, the linear scanning strategy creates a more homogeneous and larger melt pool compared to the concentric circle scanning strategy, providing more reliable support for the weld strength.
2. The experiments demonstrate that under the four operating conditions (no intermediate layer without external pressurization, NIL-NEP; Alcohol liquid film-assisted

without external pressurization, AL-NEP; No intermediate layer with external pressurization, NIL-EP; Alcohol liquid film-assisted with external pressurization, AL-EP), and the LFTW method achieves a weld strength comparable to that of the externally pressurized condition, with a maximum shear strength of 19.02 MPa. Although the shear strength under the liquid film-assisted and externally pressurized condition (AL-EP) is slightly lower than that of the externally pressurized condition (NIL-EP), the LFTW method simplifies the process, reduces equipment complexity and shows strong potential for engineering applications.

3. After cooling, the nanosecond laser welding process results in a mosaic structure composed of molten glass and stainless steel, which is considered a key characteristic of stable welded joints. EDS analysis reveals significant glass and metal elements intermixing in the joint area. When combined with 3D morphological analysis, the welded joint region can be divided into three areas: a smooth zone with high strength, a transition zone, and a fringe region affected by heat. The quality of the smooth zone formation primarily determines the weld strength.

4. The surface tension and capillarity of the liquid film significantly enhance optical contact between the materials while also preventing contact damage caused by thermal expansion. The high energy density of the nanosecond laser induces plasma formation and localized melting at the material interface. This process further fills the material gap through the Marangoni effect and recoil pressure, resulting in a densely welded joint. During welding, the recoil pressure and thermal expansion effects vaporize the liquid film in the intermediate region, reducing the risk of inhomogeneous welding through capillary action.

CRedit authorship contribution statement

Runhan Zhao: Writing – original draft, Validation, Formal analysis, Conceptualization.

Yide Li: Validation, Formal analysis, Data curation. **Jingyi Luo:** Validation, Formal analysis, Data curation. **Zhijiao Liu:** Validation, Formal analysis, Data curation. **Jiajun Fu:**

Validation, Formal analysis, Data curation. **Chao Liu:** Validation, Formal analysis, Data curation. **Huixin Wang:** Resources, Investigation. **Qinghua Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Tairui Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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